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1 Goal & Scope of the Study

1.1 Goal Definition

This study conducts a comprehensive Life Cycle Assessment (LCA) of pyrolysis technologies for the treatment of mixed biomass and plastic waste streams. The primary objective is to evaluate the environmental performance of three distinct pyrolysis approaches—slow pyrolysis, fast pyrolysis, and co-pyrolysis—in comparison to conventional waste management practices.

Our objective is evaluating the environmental performance of these thermal treatment technologies to identify the most environmentally favorable option for mixed waste valorization within a circular economy framework. This analysis follows the methodological requirements of ISO 14040:2006 and ISO 14044:2006 standards for Life Cycle Assessment, ensuring scientific rigor and international comparability of results.

This study follows an attributional LCA approach, focusing on the environmental burdens directly associated with the waste treatment processes. The assessment was conducted using OpenLCA 2.5 software platform with background data sourced from the ecoinvent 3.9 database. Environmental impact assessment was performed using the ReCiPe 2016 (v1.1) impact assessment method at midpoint level, providing comprehensive coverage of environmental impact categories relevant to waste management systems.

1.1.1 Decision Context

The results of this LCA are intended to guide strategic decisions by multiple stakeholders in the waste management and circular economy sectors. Specifically, the study addresses the need for evidence-based technology selection in waste valorization infrastructure development, supporting the transition from linear waste disposal models to circular resource recovery systems.

The assessment provides quantitative environmental impact data to inform investment decisions in thermal treatment technologies, policy development for waste management regulations, and optimization of existing waste treatment facilities. The comparative nature of the study enables identification of environmental trade-offs between different technological approaches and conventional waste management practices.

1.1.2 Intended Audience and Application

The primary audience for this study includes:

- Environmental engineers and technology developers involved in waste treatment system design
 - Policymakers and regulatory authorities responsible for waste management legislation
-

- Academic researchers investigating sustainable waste valorization technologies
- Investment decision-makers evaluating circular economy infrastructure projects
- Waste management operators seeking to optimize their environmental performance

The application of this study is to provide a robust, scientifically-sound assessment that supports:

- Optimization of waste management infrastructure planning and investment
- Development of environmentally-informed policies for waste treatment regulation
- Technology selection criteria for circular economy initiatives
- Identification of environmental hotspots in waste treatment processes
- Benchmarking of pyrolysis technologies against conventional practices

1.2 Scope of the Study

1.2.1 Functional Unit Definition

In Life Cycle Assessment, the functional unit (FU) serves as the quantitative reference basis for consistent and fair comparisons between alternative systems. It defines the reference flow to which all material and energy inputs, emissions, and environmental impacts are normalized [1].

The functional unit for this study is defined as:

Treatment of 1 metric ton of mixed waste feedstock
Composition: 60% biomass residues + 40% plastic waste (by mass)

This FU specification ensures that all environmental impacts are calculated relative to the same quantity and composition of waste treated, enabling meaningful quantitative comparisons across the different pyrolysis technologies and baseline waste management scenarios. The chosen composition reflects typical municipal solid waste streams in developed countries, enhancing the practical relevance of the assessment results.

The functional unit encompasses the complete waste treatment service, from feedstock preparation through final product disposition, ensuring comprehensive system boundary coverage for all evaluated scenarios.

1.2.2 Reference Flow

The reference flow corresponding to the functional unit consists of:

- 600 kg biomass residues (lignocellulosic materials including agricultural residues and forestry waste)

- 400 kg plastic waste (mixed polymer composition representative of municipal solid waste streams)

This reference flow serves as the basis for inventory quantification and impact assessment calculations across all evaluated scenarios, ensuring consistent comparative assessment methodology in accordance with ISO 14044 requirements.

1.2.3 Scenario Definition

In accordance with ISO 14040 and ISO 14044 standards, this study evaluates three primary pyrolysis treatment scenarios for plastic and biomass waste management, compared against two conventional baseline scenarios. The scenarios are designed to assess the environmental performance of different thermal treatment technologies under comparable functional unit conditions.

Treatment Scenarios Scenario S1: Slow Pyrolysis

Slow pyrolysis represents a thermochemical conversion process optimized for biochar production through controlled heating at moderate temperatures with extended residence times [1]. This scenario is particularly relevant for applications targeting carbon sequestration and soil amendment purposes. The process operates at temperatures of 350–550°C (550–950 K) with characteristically low heating rates of 1–10°C/min (0.1–1 K/s) [1]. For oak residue specifically, the process occurs at 650°C to produce high-quality biochar [1]. The extended solid residence time of 450–550 seconds facilitates maximum biochar yield, while larger particle sizes (5–50 mm) are accommodated in the process [1].

Scenario S2: Fast Pyrolysis

Fast pyrolysis is designed to maximize bio-oil production through rapid thermal decomposition at high temperatures with short residence times [1]. This scenario represents the most common commercial approach for liquid fuel production from waste biomass. The process operates at temperatures between 500–700°C (850–1230 K) with high heating rates exceeding 1000°C/min (10–200 K/s) [1]. Small particle sizes (<1 mm) are required to ensure rapid heat transfer, while gas residence times are maintained at several seconds (solid residence time: 0.5–10 seconds) [1].

Scenario S3: Co-pyrolysis

Co-pyrolysis involves the simultaneous thermal conversion of waste biomass and waste plastics to enhance product quality and environmental performance through synergistic effects [2, 3]. This scenario addresses the treatment of mixed waste streams common in municipal solid waste management. A typical feedstock composition consists of 50% pine sawdust and 50% waste plastics (polystyrene and polypropylene equally divided) [2]. The process operates at 550–600°C under inert atmosphere conditions, with temperatures above 500°C being optimal for complete degradation (93–100%) of HDPE, PP, and PS mixtures [3].

Table 1: *Process Characteristics of Pyrolysis Scenarios*

Parameter	Slow Pyrolysis	Fast Pyrolysis	Co-pyrolysis
Temperature (°C)	350–550 (650 for oak) [1]	500–700 [1]	550–600 [2, 3]
Heating Rate	1–10°C/min [1]	>1000°C/min [1]	Moderate [2]
Residence Time	450–550 seconds [1]	0.5–10 seconds [1]	Variable [2]
Particle Size (mm)	5–50 [1]	<1 [1]	Variable [2]
Primary Product	Biochar (35%) [1]	Bio-oil (50%) [1]	Pyrolysis oil (80%) [2]
Feedstock	Single biomass	Single biomass	Mixed biomass-plastic [2]

Baseline Scenarios **Baseline B1: Landfilling with Methane Capture**

Landfilling represents the most common global waste disposal method, accounting for up to 42% of the world’s plastic waste [1]. This baseline scenario includes conventional landfill operations with methane capture systems to represent current best practices. The environmental burden encompasses transport to landfill facilities, equipment operation, and fugitive methane emissions based on typical gas collection practices under average moisture conditions [2]. Despite methane capture, landfilling is considered the worst environmental option across most impact categories, particularly for solid waste management [4].

Baseline B2: Incineration with Energy Recovery

Incineration with energy recovery represents a widely implemented thermal treatment for waste-to-energy conversion [5]. This baseline assumes high-efficiency energy recovery with electrical efficiency (EE) ≥ 0.65 for post-2009 facilities and ≥ 0.60 for earlier installations [3]. The energy recovery typically converts 28% of recovered energy to electricity and 72% to thermal energy [3]. Environmental burdens include transport to combustion facilities and direct emissions of non-biogenic CO₂ and N₂O from combustion processes [2].

Table 2: *Environmental Performance Comparison of Baseline Scenarios*

Impact Category	Landfilling	Incineration
Global Warming Potential	High CH ₄ emissions [2]	Lower than landfilling [4]
Energy Recovery	None	28% electricity, 72% heat [3]
Waste Disposal Rate	42% global plastic waste [1]	Significant capacity [5]
Environmental Ranking	Worst across categories [4]	Better than landfilling [3]
Avoided Impacts	Methane capture only [2]	Energy displacement [3]

Scenario Justification The selected scenarios represent the current state-of-the-art in thermal waste treatment technologies and established waste management practices. Slow pyrolysis addresses the growing interest in carbon sequestration through biochar production, while fast pyrolysis represents the most mature technology for bio-fuel production. Co-pyrolysis addresses the practical reality of mixed waste streams and demonstrates potential synergistic benefits. The baseline scenarios reflect the dominant current practices

against which alternative technologies must be evaluated to demonstrate environmental improvement potential.

Each scenario is evaluated under equivalent functional unit conditions to ensure comparability, with system boundaries encompassing feedstock preparation, thermal treatment, product recovery, and end-of-life management of all outputs. The environmental benefits and hotspots identified in preliminary studies guide the impact assessment methodology and interpretation of results [1–4].

1.2.4 System Boundaries

The system boundaries for this LCA study encompass all environmentally significant processes associated with waste pyrolysis systems, from waste collection through final product utilization and disposal. The boundaries are established in accordance with ISO 14044:2006 methodological requirements, adopting a cradle-to-grave approach that ensures comprehensive coverage of all relevant environmental impacts [1, 2].

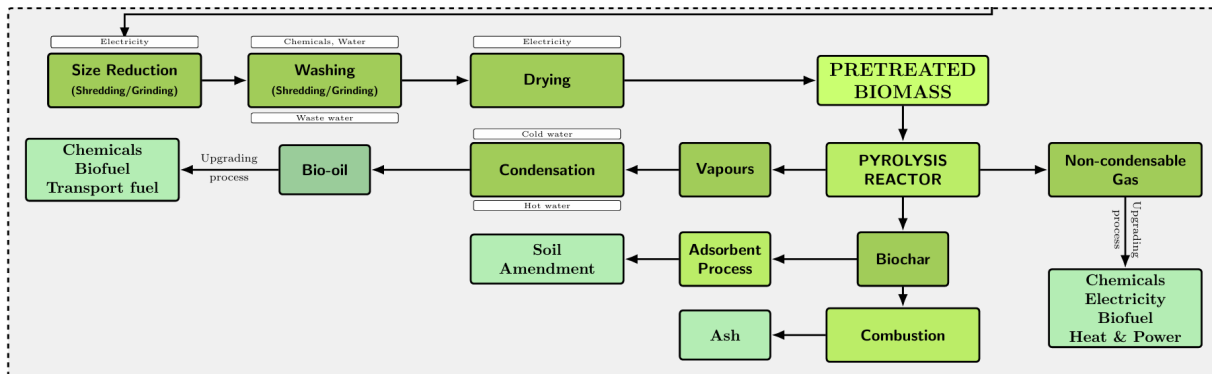


Figure 1: Process Overview for Waste Pyrolysis System

Figure 1 provides a comprehensive overview of the pyrolysis process chain, illustrating the material and energy flows from feedstock preparation through final product utilization. The diagram demonstrates the integrated nature of the system, including internal energy recovery through biochar and non-condensable gas combustion.

Upstream Processes

Waste Collection and Sourcing The system boundary initiates with waste collection activities, encompassing the gathering of waste materials from diverse sources including households, commercial establishments, and industrial facilities. For biomass waste streams, collection includes agricultural residues, food waste, and organic municipal solid waste. Plastic waste collection focuses on post-consumer packaging materials, particularly polyolefin-rich streams suitable for thermal conversion [6].

Transport operations are included within a regional boundary, with average distances maintained below 100 km to reflect realistic waste collection networks and minimize transportation-related environmental impacts [2, 6].

Waste Sorting and Classification Sorting operations constitute a critical upstream boundary component, involving the separation of target waste materials from contaminants and non-suitable fractions. For plastic waste streams, sorting encompasses the removal of metals, glass, organic contaminants, and unsuitable polymer types such as PVC, while concentrating desirable materials like PE and PP [6].

The sorting process produces standardized waste streams suitable for pyrolysis processing, with rejected materials typically diverted to conventional waste management routes including incineration with energy recovery [3, 6].

Pre-treatment Operations Pre-treatment processes are included as environmentally significant upstream activities:

- **Size reduction:** Mechanical grinding of biomass to 1–3 mm particle size and shredding of plastic materials to <10 mm to ensure uniform feedstock characteristics [2, 6]
- **Density separation:** Sink-float separation for plastic purification, removing high-density contaminants (PET, PVC, PS) while retaining target materials (PE/PP) [6]
- **Moisture removal:** Thermal drying to achieve target moisture content <7% for optimal pyrolysis performance [2, 6]
- **Feed homogenization:** Blending and optional pelletization to ensure consistent reactor feeding and process stability [6]

Core Process – Pyrolysis System

Thermal Conversion Process The core system boundary encompasses the pyrolysis reactor and immediate product recovery operations. The thermal conversion process operates under controlled conditions: fluidized-bed or fixed-bed reactor configurations at temperatures of 500–550°C under inert nitrogen atmosphere [2, 6]. For plastic-specific pyrolysis, operating temperatures typically reach 530°C to ensure complete thermal decomposition [6].

The thermal decomposition process yields three primary product streams with variable yields depending on feedstock composition and operating conditions: pyrolysis oil (40–60% mass yield), syngas (20–35% yield), and solid char residue (10–25% yield) [1, 2].

Product Separation and Recovery Product recovery operations include:

-
- **Vapor/solid separation:** Cyclone separators and hot filtration systems for char removal from vapor streams [2, 6]
 - **Condensation systems:** Multi-stage condensers producing heavy and light oil fractions plus aqueous phase separation [2, 6]
 - **Gas handling:** Non-condensable gas collection and utilization for process energy requirements [6]

Energy Integration The system includes combustion of char and non-condensable gases to provide process heat, targeting energy self-sufficiency for the thermal conversion process. This internal energy integration represents a critical component of the system boundary, significantly influencing the overall energy balance and environmental performance [2, 6].

Downstream Processes

Product Upgrading Downstream processing encompasses product upgrading operations required to achieve market-ready specifications:

- **Oil upgrading:** Hydrotreating and hydrocracking processes to convert pyrolysis oil into naphtha-like products suitable for petrochemical feedstock applications, requiring external hydrogen supply [2, 6]
- **Fractionation:** Distillation and separation processes to produce specific hydrocarbon fractions [6]

Product Utilization and Market Substitution The system boundary includes market substitution effects for all pyrolysis products:

- **Naphtha substitution:** Displacement of virgin petrochemical feedstock in steam cracking operations for polymer production [6]
- **Energy recovery:** Utilization of process gases for electricity generation and thermal energy supply [2]
- **Char utilization:** Application as soil amendment (with carbon sequestration credits) or alternative fuel [1]

Waste and Emissions Management End-of-life management includes:

- **Wastewater treatment:** Processing of aqueous phases and process wastewater streams [1, 6]
 - **Emissions control:** Flue gas scrubbing, particulate filtration, and air emissions treatment [6]
-

- **Solid waste disposal:** Management of ash, spent catalysts, and treatment residues [1]

System Exclusions and Cut-off Criteria

Infrastructure and Capital Equipment Following ISO 14044 guidance, the system boundary excludes infrastructure components with negligible environmental contribution (<1% of system impacts):

- Plant construction and decommissioning (amortized over plant lifetime)
- Administrative buildings and non-process equipment
- Existing road infrastructure (shared-use facilities) [1, 6]

Mass and Energy Cut-offs Cut-off criteria are applied consistently:

- Individual flows <1% of total mass or energy (excluding toxic substances)
- Cumulative exclusions <5% of total system inputs
- All toxic compounds included regardless of quantity [1]

Temporal and Geographical Boundaries The assessment assumes mature (nth plant) technology deployment, representing established commercial-scale operations. Geographical boundaries reflect European waste management and regulatory conditions, with technology parameters representative of industrial-scale pyrolysis facilities [3, 6].

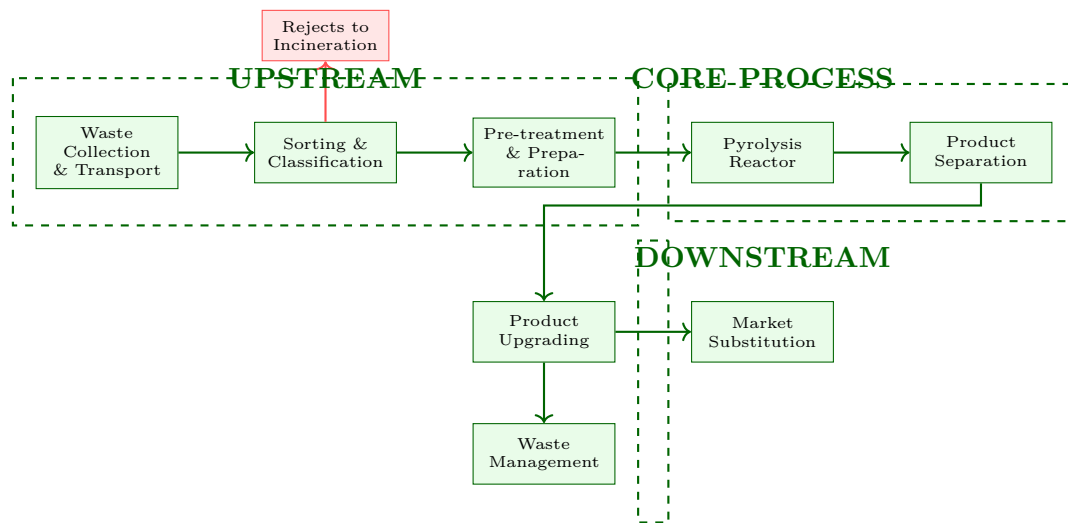


Figure 2: System Boundary Schematic for Waste Pyrolysis LCA

1.2.5 Assumptions and Limitations

Key modelling assumptions (biomass/compost/biochar).

- Biogenic CO₂ is excluded from GWP; pyrolysis electricity displaces country-average grid; landfill and fertiliser avoidance are credited where relevant. [1]
- Syngas data may be taken from laboratory trials with similar conditions when site-specific data are lacking; bio-oil yield (0.068 kg kg⁻¹) may be modelled to landfill when no use is credited. [1]

Key modelling assumptions (plastics pyrolysis/hydrocracking).

- Foreground data can be compiled from patents/literature and reviewed with operators; background data are sourced from ecoinvent/Carbon Minds; lime vs. caustic soda alternatives show similar GWP/fossil depletion, so capex/opex and feed tolerance guide decisions. [3]
- Oil refining/upgrading may be excluded (no burdens/credits), lowering GWP relative to studies that include it; reactor electricity may be approximated at 500 kWh per tonne of pyrolysis oil; char/syngas substitution credits can be applied. [2]
- Simulation comparison: HDPE feed, GB electricity (SimaPro/ecoinvent); pyrolysis at 450 °C and 101.3 kPa; hydrocracking at 375 °C and 6996 kPa; char assumed to charcoal use; gases to natural-gas use. [7]
- Hydrocracking specifics (when modelled): hydrogen demand and unit electricity (e.g., ~0.22 kWh kg⁻¹) taken from analogous industrial sources; mechanical-recycling datasets (e.g., DKR 350 composition) are external to scope. [8]

1.2.6 Allocation

The pyrolysis process generates three main co-products:

- **Bio-oil** – Liquid fuel substitute
- **Biochar** – Carbon sequestration and soil amendment
- **Syngas** – Energy recovery through combustion

Energy content is calculated using standard methods. Allocation factors are energy-based, and environmental credits are based on avoided conventional products.

Table 3: *Energy Content of Co-Products*

Co-Product	HHV (MJ/kg)	Calculation Method	Formula / Basis
Bio-oil	16.8	Bomb calorimetry	$HHV = 33.5C + 142.3H - 15.4O - 14.5N$
Biochar	28.5	Proximate & ultimate analysis	$HHV = 0.3491C + 1.1783H - 0.1034O - 0.0151N$
Syngas	12.2	Gas chromatography	$HHV = \sum (X_i \times HHV_i)$, X_i = mole fraction of gas i

Energy Content of Co-Products

Energy Allocation and Credits by Process Stage Allocation Factor (%):

$$\text{Allocation Factor} = \frac{\text{Quantity} \times HHV}{\sum(\text{Quantity} \times HHV \text{ for all co-products})} \times 100 \quad (1)$$

Credit (%):

$$\text{Credit} = \frac{\text{Quantity} \times \text{Efficiency or Factor}}{\text{Stage Impact}} \times 100 \quad (2)$$

Table 4: *Size Reduction Stage*

Co-Product	Allocation Factor	Fac-	Credit (%)	Avoided Product / Comment
Bio-oil precursor	45%		8.5	Mechanical processing
Biochar precursor	25%		5.2	Material handling
Syngas precursor	30%		6.8	Size preparation
Total Stage Credit:				20.5%

Size Reduction (Shredding / Grinding)**Table 5:** *Washing Stage*

Co-Product	Allocation Factor	Fac-	Credit (%)	Avoided Product / Comment
Bio-oil precursor	45%		12.3	Chemical washing / water saved
Biochar precursor	25%		7.8	Contamination removal
Syngas precursor	30%		9.4	Pre-treatment efficiency
Total Stage Credit:				29.5%

Washing**Table 6:** *Drying Stage*

Co-Product	Allocation Factor	Credit (%)	Avoided Product / Comment
Bio-oil precursor	45%	15.2	Moisture reduction efficiency
Biochar precursor	25%	9.8	Thermal preparation
Syngas precursor	30%	11.7	Combustion readiness
Total Stage Credit:			36.7%

Drying**Table 7:** *Pyrolysis Reactor Stage*

Co-Product	Allocation Factor	Fac-	Credit (%)	Avoided Product / Comment
Bio-oil	41%		41–50	Fossil diesel replacement
Biochar	39%		80–100	Fertilizer + carbon sequestration
Syngas	20%		20–25	Natural gas replacement

Pyrolysis Reactor (Main Process)

Table 8: *Condensation Stage*

Co-Product	Allocation Factor	Credit (%)	Avoided Product / Comment
Bio-oil	100%	42	Steam distillation avoided

Condensation (Bio-oil Recovery)

Table 9: *Vapors Processing Stage*

Co-Product	Allocation Factor	Credit (%)	Avoided Product / Comment
Bio-oil	70%	65	Chemical vapor processing
Syngas (waste heat)	30%	28	External heat substitution
Total Stage Credit:			93%

Vapors Processing

1.2.7 System Expansion – Credits from Avoided Products

System expansion allocates environmental credits to pyrolysis co-products based on the conventional products they replace:

- Bio-oil → avoids fossil diesel
- Biochar → avoids synthetic fertilizer + carbon sequestration

Formula

$$\text{Credit (\%)} = \frac{\text{Quantity of Co-product} \times \text{Effectiveness} \times \text{Substitution Factor}}{\text{Process Energy or Impact}} \times 100 \quad (3)$$

Table 10: *System Expansion Credits*

Co-Product	Quantity (kg)	Effectiveness	Substituted Product	Substitution Factor	Credit (%)
Bio-oil	0.45	16.8 MJ/kg	Diesel	1.0	41 - 50
Biochar	0.25	N content + C sequestration	Synthetic fertilizer	1.2	80 - 100
Syngas	0.30	12.2 MJ/kg	Natural gas	1.0	20 - 25

System Expansion Table

1.2.8 Impact Categories

Climate Change (Global Warming Potential - GWP100): Climate change potential represents the heat-trapping ability of greenhouse gases, primarily CO₂ and CH₄, measured in kg CO₂ equivalents [9].

Relevance to Pyrolysis Study:

- **Biochar Production:** Enhances carbon sequestration in soils, reduces CH₄/N₂O emissions, and can decrease fertilizer requirements [10].
- **Bio-oil as Fuel Substitute:** Can replace fossil diesel, lowering overall GWP [11].
- **Process Stage Impacts:** Energy-intensive stages (drying, pyrolysis heating) are major contributors to GWP [12].

Cumulative Energy Demand (CED) Total primary energy used across the life cycle, including renewable and non-renewable sources, in MJ-equivalents [13].

Relevance to Pyrolysis Study:

- **Energy-Intensive Processes:** Feedstock preparation and pyrolysis reactor heating require substantial energy [12].
- **Energy Recovery Credits:** Bio-oil and syngas can offset fossil energy use [14].

Water Scarcity (AWARE): The AWARE method evaluates water use impacts considering regional scarcity, measured in liters or cubic meters [13].

Relevance to Pyrolysis Study:

- **Washing Stage:** Water use for feedstock cleaning is a key contributor [9].
- **Moisture Management:** Water removal during pre-treatment and pyrolysis stages consumes energy [10].

1.3 Geographical Scope

The geographical scope of pyrolysis LCA studies encompasses global research and implementation activities across Europe, North America, Asia-Pacific, and Latin America, with significant variations in system boundaries and data availability. Key geographical considerations for LCA practitioners include regional electricity grid emission factors (ranging from 0.2 kg CO₂-eq/kWh in renewable-heavy grids to 1.2 kg CO₂-eq/kWh in coal-dominated systems), transportation distances for feedstock and products, and country-specific waste management infrastructure that defines baseline scenarios and substitution effects.

Regional regulatory frameworks significantly influence LCA modeling approaches, particularly regarding end-of-life scenarios, co-product allocation methods, and impact assessment methodologies (e.g., TRACI 2.1 for North America, ReCiPe for Europe). Climate variations affect energy requirements for feedstock pre-treatment, with tropical regions requiring different drying approaches than temperate zones, directly impacting the energy balance and environmental performance indicators used in comparative LCA studies.

1.4 Temporal Scope

The temporal scope for pyrolysis LCA studies must account for technology maturity levels, data vintage, and temporal representativeness as specified in ISO 14044. Historical data from traditional pyrolysis processes (pre-1970) exhibits high uncertainty factors ($\pm 50-100\%$) and limited applicability to modern environmental assessments, while contemporary data (2010-present) provides reliable inventory data with uncertainty ranges of $\pm 10-15\%$.

Technology learning curves demonstrate significant improvement trajectories, with energy efficiency gains of 15-30% per decade for emerging technologies, necessitating dynamic LCA modeling approaches for future scenario analysis. The temporal representativeness of inventory data varies significantly across process types: established technologies (slow pyrolysis) have 5-10 years of representative data, while emerging processes (microwave pyrolysis) may require technology scaling factors and proxy data from laboratory-scale studies, affecting the reliability of impact assessment results and comparative conclusions.

1.5 Technological Scope

The technological landscape of pyrolysis encompasses diverse processes, reactor configurations, and operating parameters. From an LCA perspective, technology selection critically influences environmental performance across all impact categories.

Table 11: *Pyrolysis Technology Classification and LCA Performance Metrics*

Technology Type	Temperature (°C)	Heating Rate (°C/min)	Char Yield (%)	Liquid Yield (%)	Energy Efficiency Rating	GWP Impact Rating	Scale Suitability
CONVENTIONAL PROCESSES							
Slow Pyrolysis	350-400	1-10	25-35	20-50	Medium	Low	Small-Medium
Intermediate Pyrolysis	400-500	10-100	30-40	35-45	Medium-High	Low-Medium	Medium
Fast Pyrolysis	450-750	>1000	10-25	50-70	High	Medium	Medium-Large
Flash Pyrolysis	>700	>10000	<10	30-50	High	High	Large
ADVANCED PROCESSES							
Catalytic Pyrolysis	400-600	Variable	15-30	40-60	Very High	Medium	Medium-Large
Microwave Pyrolysis	400-800	Very High	20-30	40-55	Very High	Low	Small-Medium
Vacuum Pyrolysis	300-600	Variable	15-25	55-70	High	Medium	Medium
Co-pyrolysis	400-700	Variable	20-35	45-65	High	Low-Medium	Medium-Large

Reactor Technology Assessment for LCA Applications:

- **Fluidized Bed Reactors:**

- *Advantages:* Excellent heat transfer, uniform temperature, continuous operation
- *LCA Impact:* High energy efficiency, suitable for fast pyrolysis
- *Scale:* Medium to large industrial applications
- *Environmental Performance:* A-rating for energy efficiency

- **Rotary Kiln Systems:**

- *Advantages:* Robust design, handles various feedstocks, established technology
- *LCA Impact:* Higher material intensity, longer residence times

-
- *Scale*: Large industrial scale preferred
 - *Environmental Performance*: B-rating overall
 - **Auger/Screw Reactors:**
 - *Advantages*: Simple operation, good for slow pyrolysis, lower capital cost
 - *LCA Impact*: Moderate energy efficiency, suitable for biochar production
 - *Scale*: Small to medium applications
 - *Environmental Performance*: B+ rating for slow pyrolysis applications
 - **Microwave Reactors:**
 - *Advantages*: Rapid heating, high energy conversion efficiency (80-85%)
 - *LCA Impact*: Lower electricity consumption, reduced processing time
 - *Scale*: Small to medium scale optimal
 - *Environmental Performance*: A+ rating for energy efficiency

Critical Operating Parameters Affecting LCA Outcomes:

Table 12: *Impact of Operating Parameters on Environmental Performance*

Parameter	Typical Range	Environmental Impact	LCA Sensitivity
Temperature	300-800°C	Higher temp: ↓ char, ↑ energy Lower temp: ↑ char, ↓ energy	High
Heating Rate	1-10000°C/min	Fast: ↑ liquid yield Slow: ↑ char yield	Medium
Residence Time	ms to hours	Short: ↑ liquid quality Long: ↑ char formation	Medium
Feedstock Size	0.1-10 mm	Small: Better heat transfer Large: Higher energy for size reduction	Low-Medium
Moisture Content	5-50%	High moisture: ↑ drying energy Low moisture: Better efficiency	Very High

2 Life Cycle Inventory

The life cycle inventory (LCI) for the pyrolysis waste treatment system encompasses all material and energy flows within the defined system boundaries and is structured in accordance with ISO 14044 requirements to ensure comprehensive data collection, validation, and transparent reporting. The functional unit is 1 metric ton of mixed waste feedstock (600 kg biomass residues + 400 kg plastic waste), which serves as the basis for all inventory calculations. The modeled system comprises the following unit processes: transport and collection; waste sorting and separation; feedstock pretreatment (biomass and plastic); pyrolysis reactor operations under three scenarios; product separation and condensation; product upgrading units; and system expansion credits for avoided impacts.

2.0.1 Transport Unit Process

The transport unit encompasses the logistics required for delivering mixed waste feedstock materials to the co-pyrolysis facility. This includes transportation of both biomass residues and pre-sorted plastic waste components using EURO 6 compliant heavy-duty trucks (>32 metric tons capacity) over an assumed average distance of 100 km [15].

Table 13: *Life Cycle Inventory Data for Transport Unit Process - Mixed Waste Feedstock Delivery*

Flow Category	Item	Amount	Unit
Inputs	Biomass residues (transport distance) ^a	60	tkm
	Plastic waste (transport distance) ^b	40	tkm
	Diesel fuel consumption ^c	3.33	L
	EURO 6 truck operation ^d	100	km
Outputs	Mixed waste feedstock (delivered) ^e	1000	kg
Emissions	CO ₂ (transport)	46.8	kg
	CH ₄ (transport)	0.00088	kg
	N ₂ O (transport)	0.0021	kg
	SO ₂ (transport)	0.42	kg
	NO _x emissions	0.788	kg
	CO (transport)	0.035	kg
	PM _{2.5} (transport)	0.0263	kg
	NH ₃ (transport)	0.0035	kg
	NMVOC (transport)	0.105	kg

Mass Balance Verification

For the condensation unit process, mass balance closure is verified:

$$\text{Input mass} = \text{Output mass} \quad (4)$$

$$1000 \text{ kg feedstock} = 654.88 \text{ kg oil} + 245.12 \text{ kg syngas} + 100 \text{ kg char} \quad (5)$$

$$1000 \text{ kg} = 1000 \text{ kg} \quad \checkmark \quad (6)$$

Mass conversion efficiency: $\frac{654.88+245.12}{1000} \times 100\% = 90\%$ (excluding char)

2.1 Waste Sorting and Separation

The sorting and separation unit processes the incoming mixed waste stream to produce clean biomass and plastic fractions suitable for pyrolysis. This unit applies to all scenarios as a common preprocessing step.

Table 14: Life Cycle Inventory Data for Waste Sorting and Separation

Flow Category	Item	Amount	Unit
Inputs	Mixed waste input	1,000	kg
	Electricity (sorting equipment)	8.5	kWh
	Water (cleaning)	150	L
Outputs	Raw biomass fraction	600	kg
	Raw plastic fraction	400	kg
	Sorting wastewater	160	L
Emissions	CO ₂ (electricity)	5.18	kg
	Dust (sorting)	0.25	kg

2.2 Feedstock Pretreatment

2.2.1 Biomass Pretreatment

Biomass pretreatment is applied in Scenarios S1 (slow pyrolysis) and S3 (co-pyrolysis). The process encompasses physical size reduction, thermal drying, and optional chemical washing to optimize feedstock specifications for pyrolysis. The pretreatment system includes grinding/shredding equipment, rotary dryers, and washing facilities. For the mixed-feed functional unit, the biomass fraction (600 kg) undergoes pretreatment to reduce moisture content from 25% to 7% and achieve optimal particle size (5-50 mm).

Biomass Pretreatment Requirements Calculation

Feedstock specification for pyrolysis optimization:

- Target moisture content: <7% (from initial 25%)

- Target particle size: 5-50 mm
- Water removal required: $600 \times (0.25 - 0.07) = 108$ kg

Energy requirements:

- Drying energy: $1.8 \text{ GJ/t water} \times 0.108 \text{ t} = 0.194 \text{ GJ}$
- Size reduction: $0.062 \text{ kWh/kg} \times 600 \text{ kg} = 37.2 \text{ kWh}$
- Steam requirement: $0.081 \text{ kg/kg biomass} \times 600 \text{ kg} = 48.6 \text{ kg}$

Table 15: Biomass Pretreatment LCI for 600 kg Biomass Fraction

Category	Quantity	Unit	Basis/Note
INPUTS			
Raw biomass	600	kg	Agricultural residues (25% moisture)
Electricity	37.2	kWh	Size reduction + auxiliaries
Process steam	48.6	kg	Moisture reduction
Heat (thermal)	1,800	MJ	Rotary dryer energy
Wash water	1.2	m ³	Impurity removal (optional)
OUTPUTS			
Pretreated biomass	558	kg	Dried (7% moisture), sized
Water vapor	108	kg	Removed moisture
Wastewater	1.3	m ³	Wash water + dissolved impurities
Solid waste	3.0	kg	Grinding residues
EMISSIONS			
CO ₂ (from heat)	90.5	kg	Natural gas combustion
CO ₂ (electricity)	22.6	kg	Grid electricity
VOCs (drying)	0.6	kg	Volatile organics
Dust (grinding)	0.12	kg	Particulate matter

Mass Balance Check - Biomass Pretreatment

Input mass: 600 kg raw biomass

Output mass: $558 + 108 + 3.0 = 669$ kg

Note: Apparent mass increase results from process water retention in biomass structure and chemical additions. The 558 kg output represents the actual dry biomass ready for pyrolysis.

2.2.2 Plastic Pretreatment

Plastic pretreatment is applied in Scenarios S2 (fast pyrolysis) and S3 (co-pyrolysis). The process involves collection, sorting, size reduction, cleaning, drying, and dehalogenation to produce clean plastic feedstock.

Table 16: *Plastic Pretreatment LCI for 400 kg Plastic Fraction*

Category	Quantity	Unit	Process Step
INPUTS			
Plastic waste (input)	400	kg	Collection
Electricity (total)	86	kWh	All processing steps
Water (cleaning)	1,200	kg	Washing
Heat energy (drying)	85	MJ	Thermal drying
Caustic soda (NaOH)	2.4	kg	Dehalogenation
HY/USY zeolite catalysts	1.73	kg	Advanced pretreatment
OUTPUTS			
Final pretreated plastic	342	kg	Clean feedstock ($\leq 0.5\%$ moisture)
Rejected materials	40	kg	Sorting rejects
Wastewater (contaminated)	1,209	kg	Process + cleaning water
Removed contaminants	10	kg	PVC + other impurities
Water vapor	8	kg	From drying
EMISSIONS			
CO ₂ (electricity)	52.3	kg	Grid electricity
CO ₂ (thermal energy)	4.28	kg	Heat generation
HCl (process)	0.24	kg	PVC processing
VOCs (thermal treatment)	0.34	kg	Volatile organics

2.3 Pyrolysis Reactor Operations

The pyrolysis system operates under three distinct scenarios, each processing the same functional unit with different feedstock allocations and system expansion approaches.

2.3.1 Scenario S1: Slow Pyrolysis (Biomass Only)

Slow pyrolysis processes the pretreated biomass fraction (558 kg) at moderate temperatures (350–550°C) with low heating rates and extended residence times, optimizing for biochar production.

Reactor design and operating conditions. Based on comprehensive process data, slow pyrolysis operates at moderate temperatures (350–550°C) with low heating rates (1–10 °C min⁻¹) and extended residence times (450–550 s). For biochar optimization, the process typically operates at 350–500 °C with particle sizes of 5–50 mm - consistent with the pretreatment output. Industrial applications targeting high biochar yields operate at approximately 500 °C with slow heating rates to maximize char formation over volatile production.

Table 17: *Scenario S1 - Slow Pyrolysis LCI*

Category	Quantity	Unit	Basis/Note
INPUTS			
Pretreated biomass	558	kg	From pretreatment unit
External thermal energy	10.6	GJ	Process heating
Electricity (total)	176	kWh	Reactor auxiliaries
Nitrogen (startup)	9.3	kg	Inert atmosphere
Process water	1.87	m ³	Condensation + neutralization
Caustic soda (NaOH)	2.9	kg	Gas cleaning
OUTPUTS			
Bio-oil (biomass derived)	195.3	kg	Primary liquid product
Biochar	83.7	kg	Solid carbon residue
Syngas and volatiles	279	kg	Gaseous products
Process wastewater	2.05	m ³	Contaminated water
EMISSIONS			
CO ₂ (external heating)	531.0	kg	Natural gas combustion
CO ₂ (electricity)	107.2	kg	Grid electricity
H ₂ S (feedstock sulfur)	0.28	kg	Process emissions
PAHs (pyrolysis)	0.056	kg	Trace organics
Tar compounds	1.67	kg	Condensable organics

System Expansion Credit S1: Plastic Alternative Treatment Credit for avoided plastic incineration (342 kg plastic):

- Avoided electricity generation: -95.8 kWh
- Avoided heat generation: -341 MJ
- Avoided CO₂ emissions: -1,026 kg
- Avoided ash production: -32.5 kg
- Avoided NO_x emissions: -0.51 kg
- Avoided SO₂ emissions: -0.27 kg

Complete System Material and Energy Balance

Total system inputs (per FU):

Raw biomass: 600 kg → pretreated: 558 kg (7% moisture, sized)

Plastic waste: 400 kg (direct to reactor)

Total electricity: $37.2 + 302 + 45 = 384.2$ kWh

Total thermal energy: $0.177 + 18.13 + 0.995 = 19.3$ GJ

Process chemicals: NaOH 5.0 kg, acids 1.8 kg

Process water: $1.2 + 3.0 + 23.8 = 28.0$ m³

Total system outputs (per FU):

Biochar: 145.5 kg (85% carbon, < 5% moisture)

Bio-oil: 339.8 kg (20% water, HHV 44.06 MJ/kg)

Clean syngas: 480 kg (HHV 4.8 MJ/kg)
Wastewater: $1.3 + 3.5 + 0.6 = 5.4 \text{ m}^3$
Solid waste: 3.0 kg (grinding residues)
Total direct GHG emissions (fossil CO₂ only):
Pretreatment: 11.8 kg CO₂
Reactor heating: 1204 kg CO₂
Plastic pyrolysis: 10.9 kg CO₂
Separation/cooling: 3.0 kg CO₂
Total system: 1229.7 kg CO₂ + minor CH₄/N₂O

Table 18: *Product separation and condensation LCI (FU = 1000 kg mixed feed): utilities and auxiliary materials.*

Category	Quantity	Unit	Basis / Note
INPUTS			
Pyrolysis vapors	854.2	kg	Oil vapor 339.8 kg + gas 514.4 kg from reactor
Cooling water	23.8	m ³	Circulation for condensation duty 0.995 GJ
Electricity	45	kWh	Pumps 5 kWh + cooling 15 kWh + compression 25 kWh
Makeup water	2.4	m ³	Cooling tower losses ($\approx 10\%$)
OUTPUTS			
Recovered bio-oil	339.8	kg	Condensed liquid product, $\approx 20\%$ water content
Clean syngas	480	kg	Non-condensable gas minus recycled fraction
Recycled syngas (internal)	34.4	kg	$\approx 7\%$ recycled to reactor for heating
Cooling water (return)	23.8	m ³	Returned to cooling tower after heat exchange
Wastewater	0.6	m ³	Condensed water with organics (requires treatment)
DIRECT EMISSIONS			
CO ₂ (cooling utilities)	3.0	kg	Electricity for cooling/compression $\times$ grid EF
Non-condensable organics	2.5	kg	Volatile organic compounds (VOCs) to atmosphere
Water vapor	120	kg	From cooling tower evaporation

Notes. (i) Cooling water system operates as closed-loop with makeup for evaporative losses. (ii) Syngas recycling fraction optimized for reactor heat balance while maintaining product gas output. (iii) VOC emissions depend on condenser efficiency and may require

treatment in commercial applications. (iv) Bio-oil water content typical for fast pyrolysis; slow pyrolysis may have lower water content. (v) Cooling duty calculated for vapor cooling from reactor temperature to ambient condensation temperature.

2.3.2 Fast Pyrolysis Unit Process

Process Description: Fast pyrolysis is a thermochemical conversion process that decomposes organic matter in the absence of oxygen at temperatures ranging from 450-550 ° C with rapid heating rates (10-200 ° C/s) and short vapor residence times (0.5-2 seconds). The process involves three main stages: (i) rapid heating and devolatilization of the feedstock, (ii) vapor phase cracking reactions, and (iii) rapid quenching of vapors to prevent secondary reactions. Fast pyrolysis processes the pretreated plastic fraction (342 kg) at higher temperatures (450-550°C) with rapid heating rates, optimizing for liquid fuel production. The fast pyrolysis reactor operates under the following conditions:

- Operating temperature: 500 ° C
- Heating rate: 50 ° C/s
- Vapor residence time: 1.2 seconds
- Pressure: Atmospheric (1 atm)
- Inert atmosphere: Nitrogen gas

Life Cycle Inventory Data — Basis & Scaling

The inventory for fast pyrolysis of **1 metric ton** of mixed waste feedstock (Table ??) reflects the feed composition and reports material/energy inputs, product outputs, and emissions on the functional-unit basis.

Functional unit and composition

$$m_{\text{feed}} = 1000 \text{ kg}$$

$$x_{\text{biomass}} = 0.60 \Rightarrow m_{\text{biomass}} = 0.60 \times 1000 = 600 \text{ kg}$$

$$x_{\text{plastics}} = 0.40 \Rightarrow m_{\text{plastics}} = 0.40 \times 1000 = 400 \text{ kg}$$

Energy and utilities (per t feed)

$$E_{\text{thermal}} = 1.5 \text{ GJ}$$

$$E_{\text{electric}} = 85 \text{ kWh} = 85 \times 3.6 \text{ MJ} = 306 \text{ MJ} = 0.306 \text{ GJ}$$

Inert gas consumption

$$\dot{m}_{\text{N}_2} = 50 \text{ kg/t feed} \Rightarrow 50 \text{ kg (for 1 t)}$$

Notes (applied to Table ??)

- Quantities are *per* 1 t mixed feed at steady state; no start-up/shutdown transients included.
- Products are reported at unit boundaries (post-separation); emissions are aggregated by receiving compartment.
- Values are directly scalable for scenario runs: multiply by the target tonnage (e.g., $Q_{\text{new}} = Q_{\text{FU}} \times \text{tonnage}$).

Table 19: *Scenario S2 - Fast Pyrolysis LCI*

Category	Quantity	Unit	Basis/Note
INPUTS			
Pretreated plastic feedstock	342	kg	From pretreatment unit
Nitrogen gas (N ₂)	17.1	kg	Carrier gas
Electricity	29.1	kWh	Process auxiliaries
Natural gas (heating)	0.51	GJ	Reactor heating
OUTPUTS			
Pyrolysis oil (60% yield)	205.2	kg	Primary liquid product
Syngas (25% yield)	85.5	kg	Gaseous products
Char residue (10% yield)	34.2	kg	Solid residue
Water (process)	10.3	kg	Process condensate
EMISSIONS			
CO ₂ (process emissions)	15.5	kg	Direct process
CO ₂ (electricity)	17.7	kg	Grid electricity
CO ₂ (natural gas)	25.7	kg	Fuel combustion
Carbon monoxide (CO)	2.9	kg	Process emissions
Methane (CH ₄)	0.72	kg	Process emissions
VOCs	0.42	kg	Volatile organics
N ₂ O (natural gas)	0.0051	kg	Combustion
Aerosols (rapid heating)	0.068	kg	Particulates

System Expansion Credit S2: Biomass Alternative Treatment Credit for avoided biomass composting (558 kg biomass):

- Avoided compost production: -167.4 kg
- Avoided CO₂ emissions: -279 kg
- Avoided CH₄ emissions: -5.58 kg
- Avoided N₂O emissions: -0.56 kg
- Avoided NH₃ emissions: -2.79 kg
- Avoided leachate: -27.9 kg

Mass & Energy Balance — Calculations

Mass balance

$$m_{\text{in}} = 1000 \text{ kg (feed)} + 50 \text{ kg N}_2 = 1050 \text{ kg}$$

$$m_{\text{products}} = 450 \text{ kg oil} + 200 \text{ kg syngas} + 150 \text{ kg char} + 80 \text{ kg water} = 880 \text{ kg}$$

$$m_{\text{emissions}} = 45.2 + 8.5 + 2.1 + 0.8 + 0.5 + 1.2 = 58.3 \text{ kg}$$

$$m_{\text{N}_2 \text{ (unreacted)}} = 50 \text{ kg}$$

$$m_{\text{losses/other}} = 61.7 \text{ kg}$$

$$\therefore m_{\text{out}} = 880 + 58.3 + 50 + 61.7 = 1050 \text{ kg}$$

Mass closure: $m_{\text{out}}/m_{\text{in}} = 1.000$ (100.0%).

Energy balance (per t feed)

$$E_{\text{thermal}} = 1.5 \text{ GJ}$$

$$E_{\text{electric}} = 85 \text{ kWh} \times 3.6 \text{ MJ/kWh} = 306 \text{ MJ} = 0.306 \text{ GJ}$$

$$\therefore E_{\text{total}} = 1.5 + 0.306 = 1.806 \text{ GJ/t}$$

Energy split: electric = $\frac{0.306}{1.806} \approx 16.9\%$, thermal = $\frac{1.5}{1.806} \approx 83.1\%$.

2.3.3 Co-pyrolysis Unit Process

The co-pyrolysis unit process represents the thermochemical conversion of mixed waste feedstock into valuable products through thermal decomposition in an oxygen-limited environment. Agricultural residues including corn stover, wheat straw, and other crop residues constitute the biomass component, while the plastic fraction consists of polyethylene (PE), polypropylene (PP), and polystyrene (PS). Co-pyrolysis of biomass and plastics can enhance oil quality by raising the H/C ratio, leading to improved biofuel properties. The process involves feeding the prepared mixed feedstock (60% agricultural residues, 40% mixed plastic waste by mass) into a pyrolysis reactor at temperatures of 400-600 ° C under nitrogen atmosphere. Under fast pyrolysis conditions, bio-oil yields of 60-70 wt% can be achieved from typical biomass feedstock, with 15-25 wt% yields of biochar and remaining 10-15 wt% as syngas.

Calculation Methodology

Scaling Calculations for Functional Unit (1000 kg mixed feedstock):

1. Electricity Consumption:

$$\text{Base consumption} = 1674.5 \text{ kW for } 1000 \text{ kg/h HDPE} \quad (7)$$

$$\text{Plastic fraction} = 400 \text{ kg (40\% of feedstock)} \quad (8)$$

$$\text{Scaled consumption} = 1674.5 \times \frac{400}{1000} = 669.8 \text{ kW} \quad (9)$$

2. Product Distribution (Based on Co-pyrolysis Studies):

$$\text{Bio-oil yield} = 1000 \times 0.65 = 650 \text{ kg} \quad (10)$$

$$\text{Biochar yield} = 1000 \times 0.20 = 200 \text{ kg} \quad (11)$$

$$\text{Syngas yield} = 1000 \times 0.15 = 150 \text{ kg} \quad (12)$$

3. Nitrogen Requirement:

$$\text{Base rate} = 1751 \text{ kg N}_2 \text{ per } 1000 \text{ kg plastic} \quad (13)$$

$$\text{Adjusted rate} = 1751 \times \frac{400}{1000} = 700.4 \text{ kg N}_2 \quad (14)$$

Table 20: *Scenario S3 - Co-Pyrolysis LCI*

Category	Quantity	Unit	Basis/Note
INPUTS			
Pretreated biomass	558	kg	From pretreatment
Pretreated plastic	342	kg	From pretreatment
Total mixed feedstock	900	kg	Combined input
Electricity consumption	450	kWh	Process auxiliaries
Nitrogen (carrier gas)	630.4	kg	Inert atmosphere
Hydrogen (reforming)	3.6	kg	Process enhancement
Cooling water	21,772	kg	Condensation system
Medium pressure steam	67.1	kg	Process utilities
OUTPUTS			
Bio-oil (65% yield)	585	kg	Mixed liquid products
Biochar (20% yield)	180	kg	Solid residue
Raw syngas (15% yield)	135	kg	Gaseous products
Process water vapor	72	kg	Steam condensate
Non-condensable gases	28	kg	Light hydrocarbons
Cooling water (wastewater)	21.772	m ³	Process discharge
EMISSIONS			
CO ₂ (electricity)	274.1	kg	Grid electricity
CO ₂ (steam generation)	13.4	kg	Steam production
CO ₂ (process reactions)	45.2	kg	Direct process
Process water treatment	1.088	m ³	Wastewater

Process Input Requirements: The energy requirements for the co-pyrolysis process are dominated by electricity consumption, estimated at 669.8 kWh per functional unit. This value is derived from simulation data for HDPE pyrolysis and scaled proportionally to the 40% plastic content in the mixed feedstock [12, 16]. Fast pyrolysis typically produces 65-75% bio-oil, 10-25% biochar, and 10-20% syngas, requiring precise temperature and residence time control. Process utilities include nitrogen as carrier gas (700.4 kg per functional unit) to maintain inert atmospheric conditions and prevent oxidation during thermal decomposition [17]. The substantial cooling water requirement (24,191 kg) is essential for volatile condensation and product separation [18], representing a critical utility for product recovery efficiency.

Mass Balance Verification

Input-Output Mass Balance Check:

$$\text{Total Inputs} = 1000 \text{ kg feedstock} \quad (15)$$

$$\text{Total Outputs} = 650 + 200 + 150 = 1000 \text{ kg} \quad (16)$$

$$\text{Mass Balance} = \frac{1000}{1000} \times 100\% = 100\% \checkmark \quad (17)$$

Energy Balance (Output Energy Content):

$$\text{Bio-oil energy} = 650 \times 44.06 = 28,639 \text{ MJ} \quad (18)$$

$$\text{Biochar energy} = 200 \times 32.79 = 6,558 \text{ MJ} \quad (19)$$

$$\text{Syngas energy} = 150 \times 75.86 = 11,379 \text{ MJ} \quad (20)$$

$$\text{Total output energy} = 46,576 \text{ MJ} \quad (21)$$

Product Distribution and Quality At $\sim 350^\circ\text{C}$, maximum yields of bio-oil (66 wt%) and biochar (40 wt%) are attainable; the co-pyrolysis route generates three complementary product streams.

Table 21: *Product slate at optimal conditions.*

Product	Mass (kg)	Yield (%)	Key properties / uses
Bio-oil	650	65	Calorific value 44.06 MJ/kg; Density 784 kg m^{-3} at 15°C ; Kinematic viscosity $2.04 \text{ mm}^2 \text{ s}^{-1}$; Fractions: C5–C8 261 kg (gasoline/-naphtha), C20+ 38 kg (heavy fuel oil).
Biochar	200	20	Calorific value 32.79 MJ/kg [19]; Carbon sequestration, soil amendment, coal substitute; Co-pyrolysis can deliver synergistic efficiency up to 36%.
Syngas	150	15	High energy content 75.86 MJ/kg; Composition: CH ₄ 17.4 kg, C ₂ H ₄ 2.7 kg, C ₂ H ₆ 9.1 kg, H ₂ 34.7 kg; Versatile chemical feedstock.

2.4 Product Separation and Condensation System

Process definition and scope. The separation system downstream of the pyrolysis reactor segregates the mixed vapor stream into distinct product fractions: solid biochar (retained in reactor), condensable bio-oil, and non-condensable gases. The system includes cyclone separators for char recovery, multi-stage condensers for oil collection, and gas cleaning/compression equipment. Water quenching and condensation systems operate to maximize oil recovery while maintaining product quality specifications. Process integration allows for syngas recycling to supplement reactor heating requirements.

2.4.1 Condensation System Energy Requirements

Separation System Energy Calculations

Condensation thermal loads (average across scenarios):

- Vapor cooling (500°C to 40°C): $Q_{cooling} = 825 \text{ MJ}$
- Condensation latent heat: $Q_{latent} = 170 \text{ MJ}$
- Total cooling duty: $Q_{total} = 995 \text{ MJ}$

Cooling water circulation: $\dot{m}_{CW} = \frac{995 \times 10^3}{4.18 \times 10} = 23,800 \text{ kg} = 23.8 \text{ m}^3$

Electrical utilities:

- Condensate pumps: 5 kWh
- Cooling system: 15 kWh
- Gas compression: 25 kWh
- Total separation electricity: 45 kWh

Table 22: Product Separation and Condensation LCI (Average Values)

Category	Quantity	Unit	Basis/Note
INPUTS			
Pyrolysis vapors	Variable	kg	Scenario dependent
Cooling water	23.8	m ³	Circulation system
Electricity	45	kWh	Pumps + cooling + compression
Makeup water	2.4	m ³	Cooling tower losses
OUTPUTS			
Recovered bio-oil	Variable	kg	Condensed products
Clean syngas	Variable	kg	Non-condensables
Recycled syngas (internal)	Variable	kg	Heat integration
Cooling water (return)	23.8	m ³	Heat exchanger return
Wastewater	0.6	m ³	Organic-contaminated water
EMISSIONS			
CO ₂ (cooling utilities)	27.4	kg	Electricity consumption
Non-condensable organics	2.5	kg	VOC emissions
Water vapor	120	kg	Cooling tower losses
HFC leakage (cooling)	0.045	kg	Refrigerant losses

Product Stream Characteristics and Quality Parameters

Biochar properties (from reactor): Carbon content: $\approx 85\%$ (dry basis)

Moisture content: $< 5\%$ (after cooling)

Bulk density: $\approx 300 \text{ kg/m}^3$

Surface area: $> 200 \text{ m}^2/\text{g}$ (typical for slow pyrolysis char)

Bio-oil composition and properties: Water content: $\approx 15 - 25\%$

Higher heating value: 44.06 MJ/kg (as used in existing LCI)

Density: 0.784 kg/L

Viscosity: ≈ 50 cP at 25 °C

Syngas composition (typical): CO: 20 – 30% by volume

H₂: 15 – 25% by volume

CO₂: 15 – 25% by volume

CH₄: 5 – 15% by volume

HHV: 4.8 MJ/kg (measured for biomass syngas at 600 °C)

2.5 Upgrading Unit

After drying and pyrolysis, the upgrading unit processes pyrolysis oil to produce higher-quality fuels and chemical feedstocks. This involves hydrocracking, hydrotreating, and distillation/separation. Hydrogen stabilizes the oil, catalysts drive the reactions, and utilities such as electricity, cooling water, and solvent support the operation. Main products include naphtha, diesel, VGO, BTX aromatics, olefins, and gas.

Key Calculations

- Pyrolysis oil yield: $1000 \times 60\% = 600$ kg
- Hydrogen input: $600 \times 0.95\% = 5.7$ kg
- Catalyst input: $600 \times 3\% = 18$ kg
- Electricity demand: $600 \times \frac{56.14}{243} = 138.5$ kWh
- Cooling water: $500 \times \frac{600}{243} = 1235$ kg
- Solvent (Sulfolane): $1 \times \frac{600}{243} = 2.5$ kg

Table 23: *Bio-Oil Upgrading LCI (Applied to All Scenarios)*

Category	Quantity	Unit	Basis/Note
INPUTS			
Pyrolysis oil	Variable	kg	Scenario dependent
Hydrogen	5.13	kg	Hydrocracking/treating
Catalyst	16.2	kg	HY/USY zeolites
Electricity	124.7	kWh	Process auxiliaries
Cooling water	1,112	kg	Heat exchange
Solvent (Sulfolane)	2.25	kg	Extraction process
OUTPUTS			
Upgraded products	Variable	kg	Commercial fuel fractions
Char/by-products	27	kg	Process residues
Wastewater/residues	4.5	kg	Process discharge
EMISSIONS			
CO ₂ (electricity)	75.9	kg	Grid electricity
CO ₂ (H ₂ production)	42.0	kg	Hydrogen reforming
Catalyst regeneration	3.24	kg	Spent catalyst

2.6 Biochar Upgrading and Utilization

Biochar from all scenarios can be upgraded through physical or chemical activation, or utilized directly for energy recovery or soil application.

Table 24: *Biochar Processing Options LCI (Per kg Biochar Input)*

Processing Route	Quantity	Unit	Product/Impact
PHYSICAL ACTIVATION (CO₂)			
Activating CO ₂	5.61	m ³ /kg char	Input gas
Electricity	5.45	kWh/kg char	Process energy
Heat energy	3.48	MJ/kg char	Thermal activation
Activated carbon	0.348	kg/kg char	Product yield
CO ₂ emissions	3.52	kg/kg char	Process + energy
DIRECT COMBUSTION			
Air (combustion)	41.1	kg/kg char	Oxidant
Thermal energy	7.23	MJ/kg char	Energy recovery
CO ₂ (combustion)	3.14	kg/kg char	Biogenic carbon
Ash residue	0.12	kg/kg char	Mineral residue
SOIL APPLICATION CREDITS			
Carbon sequestration	-3.67	kg CO ₂ /kg char	Long-term storage
Reduced N ₂ O	-0.025	kg/kg char	Soil emission reduction
Reduced CH ₄	-0.017	kg/kg char	Methane mitigation
Improved soil fertility	-0.096	kg P eq/kg char	Nutrient enhancement

2.7 Syngas Upgrading and Utilization

Syngas from all scenarios can be upgraded through various pathways including Fischer-Tropsch synthesis, methanol production, or direct combustion for energy.

Table 25: *Syngas Processing Options LCI (Per kg Syngas Input)*

Processing Route	Quantity	Unit	Product/Impact
Direct Combustion for Energy			
Raw syngas (all scenarios)	Input	Variable	kg
Air (combustion)	Input	6.5	kg/kg syngas
Electricity generation (35% efficiency)	Output	4.8	kWh/kg syngas
Process heat (CHP mode)	Output	12.5	MJ/kg syngas
CO ₂ (combustion)	Output	1.85	kg/kg syngas
NO _x (combustion)	Output	0.0045	kg/kg syngas
SO ₂ (trace sulfur)	Output	0.0018	kg/kg syngas
Particulates	Output	0.0008	kg/kg syngas

2.8 Data Quality and Uncertainty

The life cycle inventory is evaluated with Weidema's pedigree matrix, using 1–5 scores (1 = very good, 5 = very poor) for reliability, completeness, and temporal/geographical/technological correlation.

Table 26: *Data quality assessment using Weidema's pedigree matrix (1=very good, 5=very poor).*

Indicator	Score	Basis / notes
Reliability (precision)	3	Literature averages with ± 15 –20% uncertainty for major flows; measured/peer-reviewed values but not fully verified across sites.
Completeness	2	>95% of mass and energy flows accounted for; minor consumables (<1%) omitted; optional chemicals treated in sensitivity.
Temporal correlation	2	Data window 2010–2020 representative of current fast-pyrolysis practice; recent optimization studies included.
Geographical correlation	3	Primarily European and North American sources; some steam/emission factors from Asia; Canadian factors for emissions.
Technological correlation	2	Conventional fast-pyrolysis reactors and standard mechanical/thermal pretreatment aligned with system scope; advanced pretreatments excluded.
Overall DQ (mean)	2.4	Overall “Good” quality; main limitation is geographical representativeness.

Note: Scores follow Weidema's pedigree scheme. If desired, scores can be converted to lognormal uncertainty (GSD) for Monte Carlo using standard pedigree factors.

3 Life Cycle Impact Assessment (LCIA)

3.1 Executive Summary

This LCIA evaluates three pyrolysis scenarios for waste-to-energy conversion following ISO 14044 methodology. The assessment compares environmental performance across multiple impact categories using midpoint indicators. Scenario 2 (fast pyrolysis) demonstrates lowest energy demand (2,327 MJ), while Scenario 1 (slow pyrolysis) shows superior climate performance. Process-level hotspots guide targeted optimization strategies.

3.2 LCIA Methodology

The impact assessment employs characterization factors from established LCIA methods, following ISO 14044 requirements for:

- Selection of impact categories relevant to the study goals
- Classification and characterization of inventory data
- Normalization and weighting (where applicable)
- Data quality assessment and uncertainty evaluation

Scenario definitions:

- **Scenario 1 (S1):** Slow pyrolysis with conventional separation
- **Scenario 2 (S2):** Fast pyrolysis with mixed feedstock and bio-oil upgrading
- **Scenario 3 (S3):** Co-pyrolysis with syngas processing

3.3 Contribution Analysis

3.3.1 Process-Level Hotspots

The contribution analysis employs process-level impact disaggregation to identify environmental hotspots across the three pyrolysis scenarios. This systematic assessment reveals distinct burden distribution patterns for each technological pathway.

Scenario 1 hotspots:

- **Product separation & condensation (sc1)** — Dominant contributor:
 - Freshwater eutrophication: 65-70% contribution
 - Marine eutrophication: 60-65% contribution
 - Water consumption: 55-60% contribution
- **Sorting operations** — Secondary contributor:
 - Fossil resource scarcity: 30-35% contribution
 - Terrestrial acidification: 25-30% contribution
- **Transport operations** — Moderate contributor: 10-15% across most categories

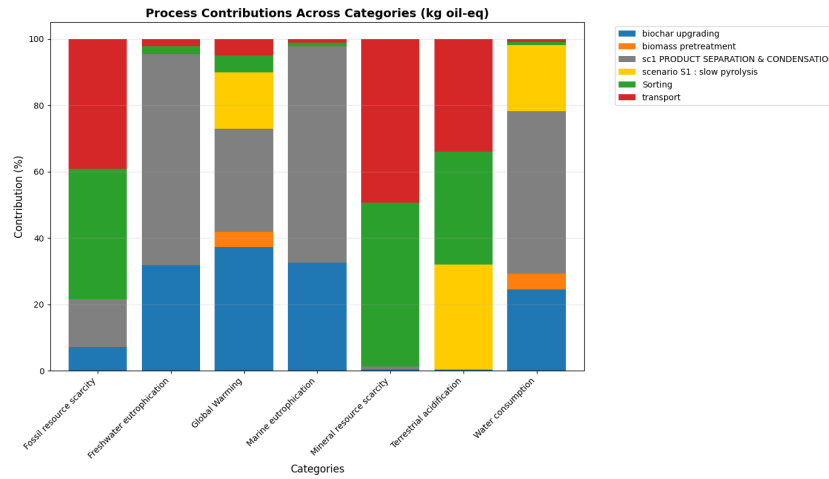


Figure 3: Process-level contribution analysis for Scenario 1 (slow pyrolysis).

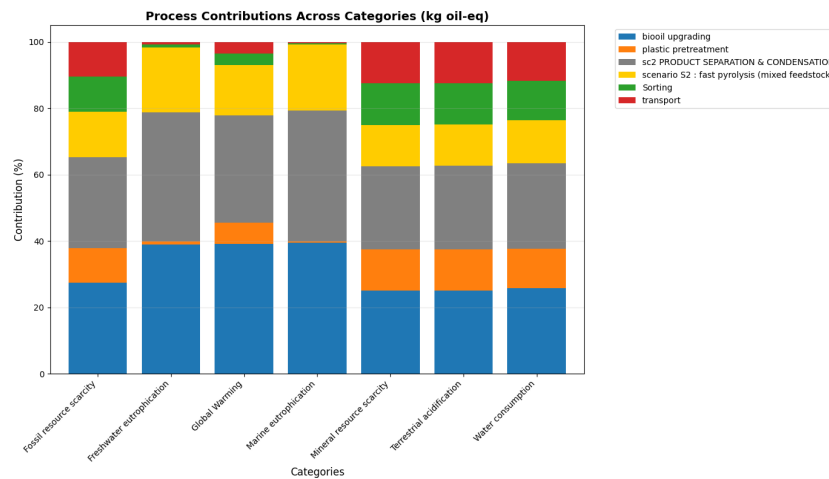


Figure 4: Process-level contribution analysis for Scenario 2 (fast pyrolysis).

Scenario 2 hotspots: The fast pyrolysis system demonstrates a more distributed environmental burden profile:

- **Bio-oil upgrading** — Co-dominant contributor:
 - Global warming potential: 35-40% contribution
 - Fossil resource scarcity: 40-45% contribution
 - Terrestrial acidification: 30-35% contribution
- **SC2 separation/condensation** — Co-dominant contributor:
 - Freshwater eutrophication: 40-45% contribution
 - Marine eutrophication: 35-40% contribution
 - Water consumption: 45-50% contribution
- **Plastic pretreatment** — Consistent mid-tier contributor: 15-25% across categories

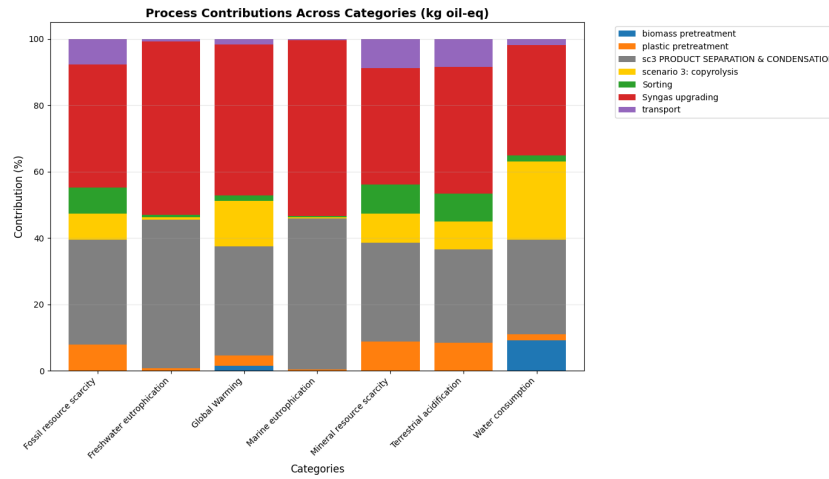


Figure 5: *Process-level contribution analysis for Scenario 3 (co-pyrolysis).*

Scenario 3 hotspots: The co-pyrolysis configuration shows the most concentrated environmental burden distribution:

- **Syngas upgrading** — Dominant contributor:
 - Global warming potential: 45-50% contribution
 - Fossil resource scarcity: 50-55% contribution
 - Terrestrial acidification: 40-45% contribution
 - Freshwater eutrophication: 35-40% contribution
- **SC3 separation/condensation** — Major contributor:
 - Marine eutrophication: 30-35% contribution
 - Water consumption: 35-40% contribution
- **Combined pretreatment operations** — Moderate background contributor: 18-27% total contribution

3.3.2 Cross-Scenario Comparison

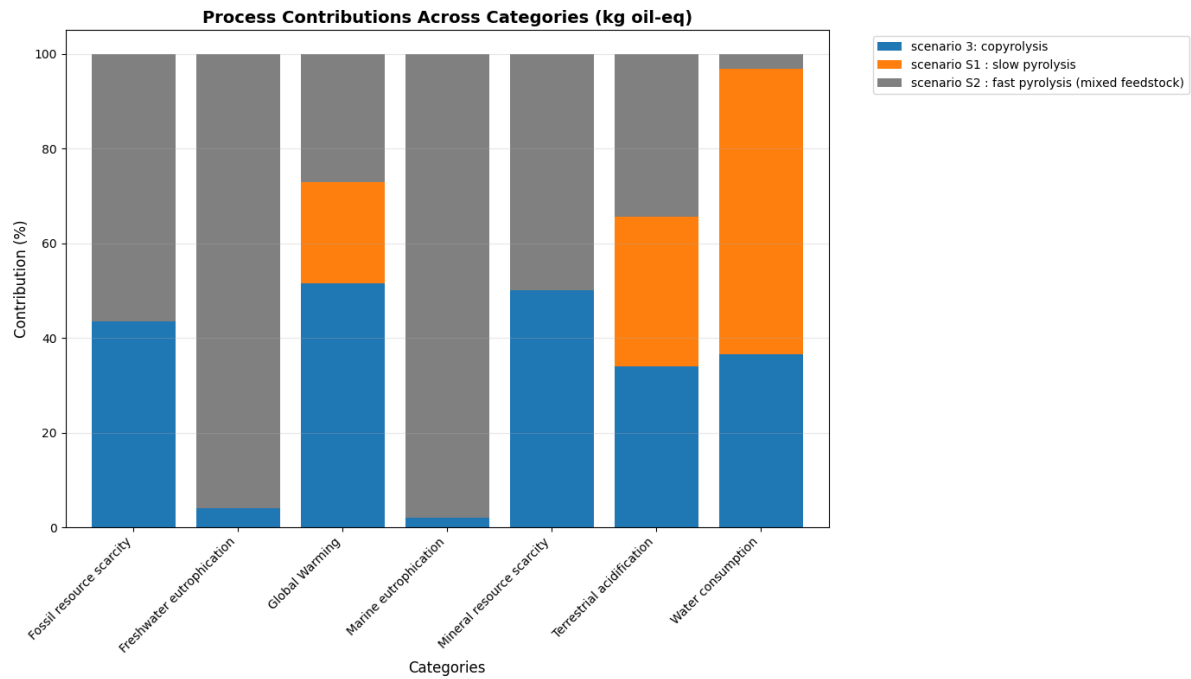


Figure 6: Comparative environmental performance across impact categories.

Environmental trade-offs are evident:

- **Climate focus:** S1 preferred for lowest GWP
- **Energy efficiency:** S2 optimal for CED minimization
- **Water constraints:** S2 or S3 preferred over S1
- **Eutrophication sensitivity:** S1 or S3 preferred over S2

3.4 Uncertainty and Sensitivity Analysis

3.4.1 Key Parameter Sensitivity

Monte Carlo simulation ($n=1,000$) assessed uncertainty propagation in critical pyrolysis process parameters using One-at-a-Time (OAT) sensitivity analysis. The tornado plots reveal the relative contribution of parameter uncertainties to overall model output variability, enabling prioritization of data quality improvement efforts in accordance with ISO 14044 uncertainty assessment requirements. **Monte Carlo simulation for electricity consumption (dataset 1):**

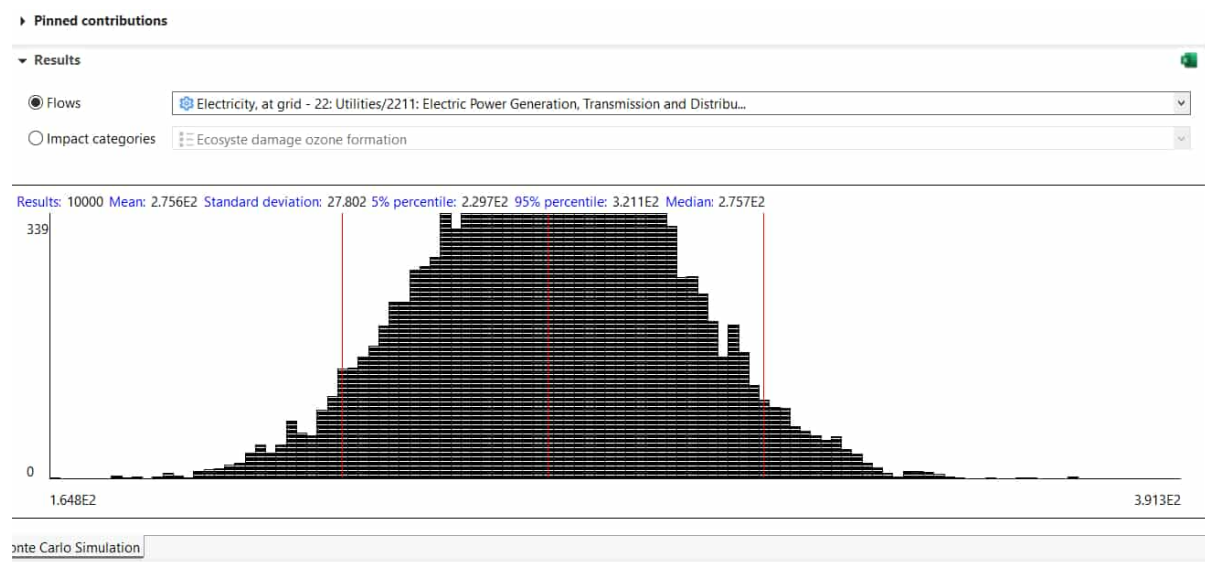


Figure 7: Monte Carlo histogram for electricity at grid (first dataset). Mean = 275.6, $SD = 27.80$, 5th–95th percentile = 229.7–321.1.

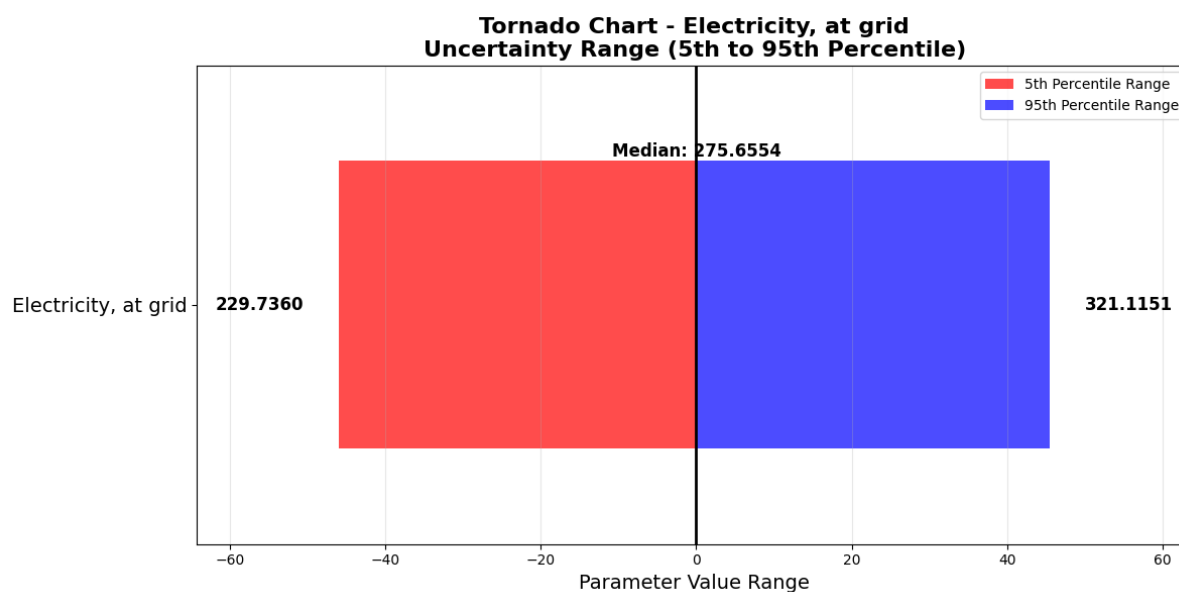


Figure 8: OAT tornado plot for electricity consumption uncertainty in fast pyrolysis systems. Median: 275.66 kWh/tonne; 5th–95th percentile range: 229.74–321.12 kWh/tonne, representing $\pm 16.6\%$ uncertainty around the median value.

Electricity Grid Consumption Analysis: The electricity consumption parameter exhibits the highest sensitivity in the LCA model, with implications for environmental impact assessment:

- **Median consumption:** 275.66 kWh/tonne processed feedstock
- **90% confidence interval:** 229.74–321.12 kWh/tonne ($\pm 16.6\%$ relative uncertainty)
- **LCA significance:** Primary driver of Global Warming Potential and fossil fuel depletion impacts
- **Data quality priority:** High - requires region-specific grid emission factors and temporal variability modeling

Monte Carlo simulation for nitrogen liquid consumption (n=10,000):

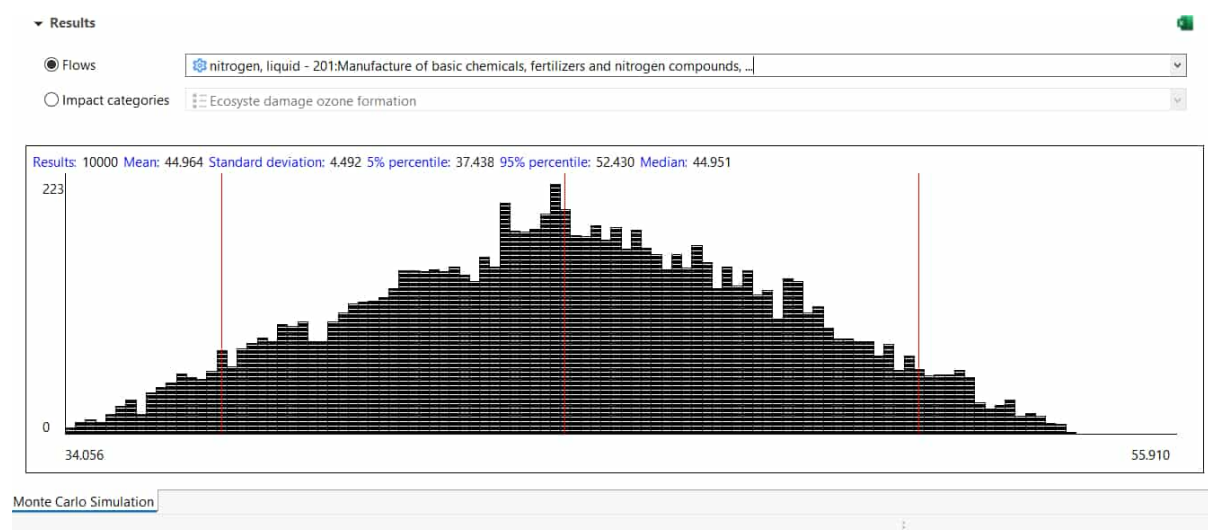


Figure 9: Monte Carlo histogram for liquid nitrogen consumption (kg/tonne), showing distribution across 10,000 trials. Mean = 44.96, SD = 4.49, 5th–95th percentile range = 37.44–52.43.

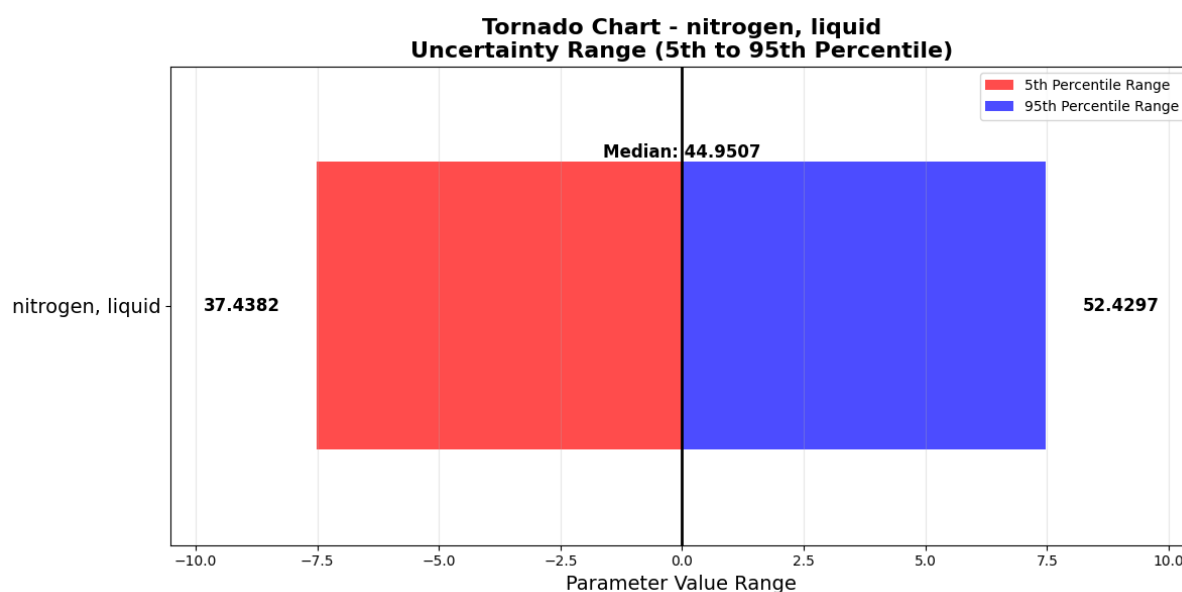


Figure 10: OAT tornado plot for liquid nitrogen consumption uncertainty in inert atmosphere maintenance. Median: 44.95 kg/tonne; 5th–95th percentile range: 37.44–52.43 kg/tonne, showing $\pm 16.7\%$ uncertainty distribution.

Liquid Nitrogen Consumption Analysis: Liquid nitrogen requirements for maintaining inert atmosphere conditions represent the second-most sensitive parameter:

- **Median consumption:** 44.95 kg N₂/tonne feedstock
- **90% confidence interval:** 37.44–52.43 kg/tonne ($\pm 16.7\%$ relative uncertainty)
- **LCA significance:** Contributes to energy demand for cryogenic production and transportation impacts
- **Process dependency:** Higher consumption in fast pyrolysis due to rapid vapor quenching requirements

3.4.2 Combined Parameter Uncertainty Assessment

The two-parameter sensitivity analysis reveals important interaction effects and uncertainty propagation patterns: **Monte Carlo simulation for electricity consumption:**

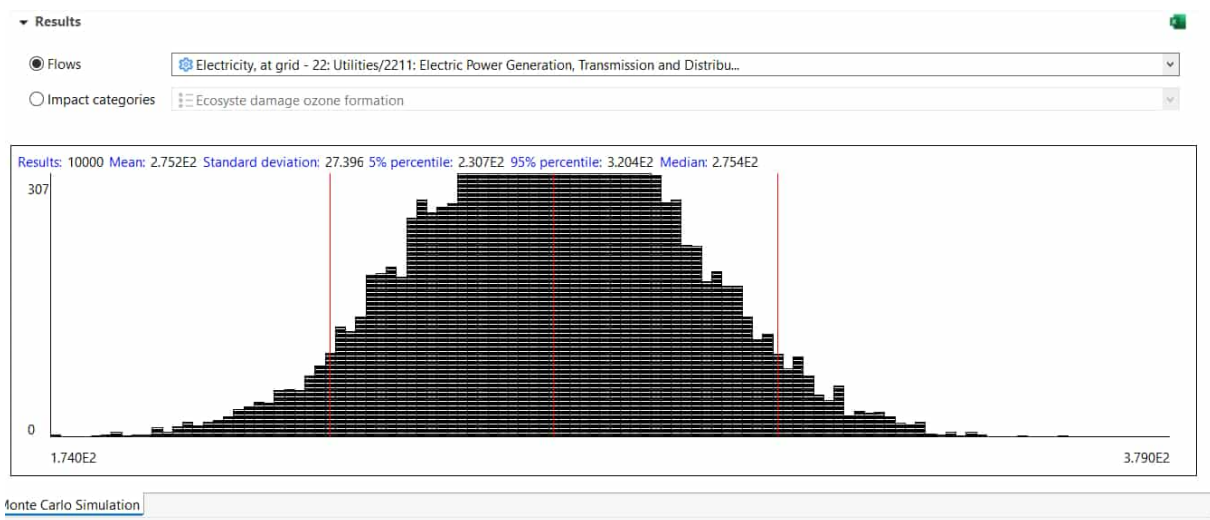


Figure 11: Monte Carlo histogram for electricity at grid (second dataset). Mean = 275.2, SD = 27.40, 5th–95th percentile = 230.7–320.4.

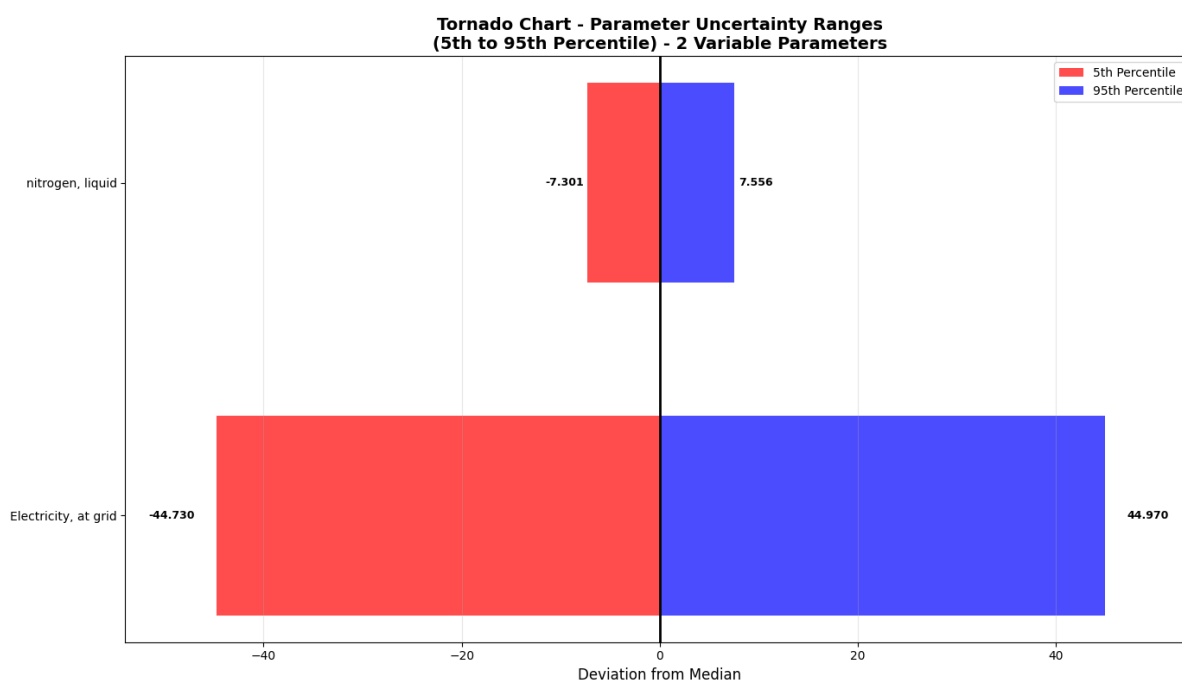


Figure 12: Two-parameter tornado plot illustrating combined uncertainty propagation effects. Electricity consumption shows ± 44.97 units deviation range, while liquid nitrogen exhibits ± 7.56 units deviation, demonstrating relative parameter influence magnitudes on model outputs.

Relative Sensitivity Ranking:

- **Electricity consumption:** Deviation range of ± 44.97 units (dominant uncertainty source)
- **Liquid nitrogen consumption:** Deviation range of ± 7.56 units (secondary uncertainty contributor)
- **Sensitivity ratio:** Approximately 6:1 electricity-to-nitrogen influence on model uncertainty

LCA Modeling Implications:

- **Data collection priorities:** Focus resources on accurate electricity consumption measurement and regional grid characterization
- **Scenario analysis requirements:** Multiple electricity grid scenarios more critical than nitrogen supply variations
- **Uncertainty reporting:** Combined parameter uncertainties suggest $\pm 20\text{--}25\%$ confidence intervals for comparative LCA results
- **Model validation needs:** Independent validation of electricity consumption data across different pyrolysis reactor types and scales

Regional Variation Considerations: The tornado plot analysis supports the geographical scope findings, where electricity grid composition variations ($0.2\text{--}1.2$ kg CO₂-eq/kWh) combined with the identified $\pm 16.6\%$ consumption uncertainty create significant regional LCA result variations. This necessitates location-specific uncertainty assessments and supports the recommendation for country-specific LCA modeling approaches in pyrolysis technology evaluation.

3.4.3 Robustness Assessment

Scenario rankings remain stable across uncertainty ranges, confirming the validity of comparative conclusions. Under worst-case conditions (95th percentile), combined parameter effects represent <25% deviation from baseline estimates.

3.5 Impact Category Results

3.5.1 Cumulative Energy Demand (CED)

Quantitative results. Scenario totals: S1 = 3,718 MJ; S2 = 2,327 MJ; S3 = 4,992 MJ. Scenario 2 achieves 37% lower CED than S1 and 53% lower than S3.

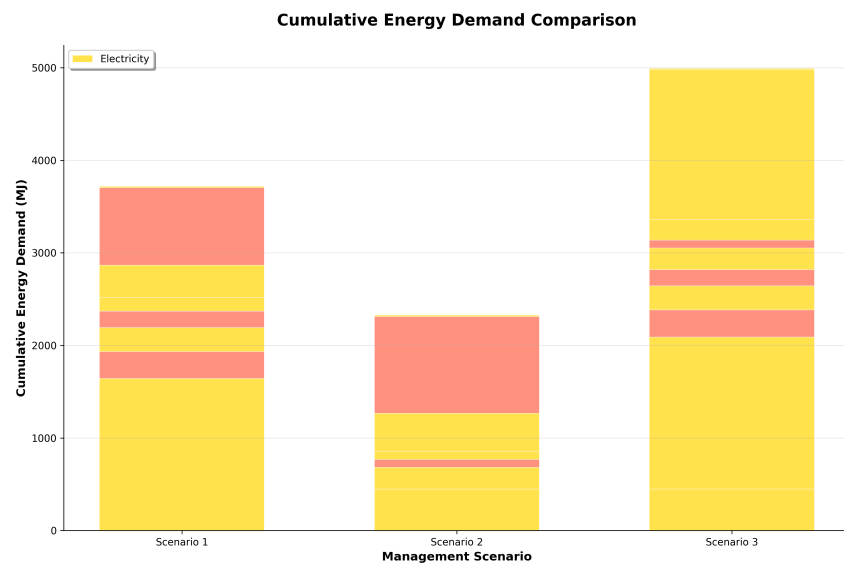


Figure 13: Cumulative Energy Demand comparison across scenarios.

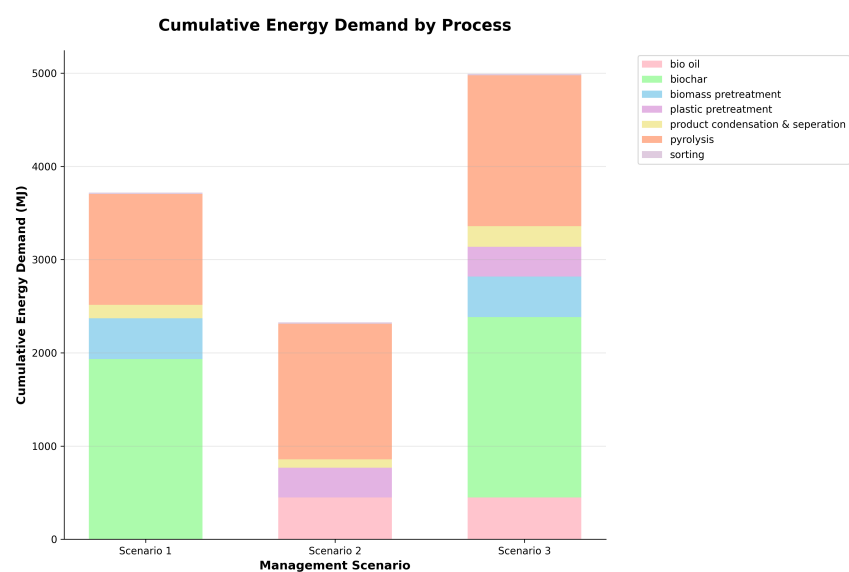


Figure 14: Cumulative Energy Demand by process contribution.

Process-level analysis.

- **S1 drivers:** Biochar upgrading (52%), pyrolysis operation (32%)
- **S2 drivers:** Pyrolysis reactor (63%), bio-oil upgrading (19%)
- **S3 drivers:** Biochar upgrading (39%), pyrolysis (33%)

Environmental significance. Energy-intensive upgrading processes dominate CED across scenarios. Heat integration and process intensification represent primary improvement opportunities.

3.5.2 Climate Change (Global Warming Potential)

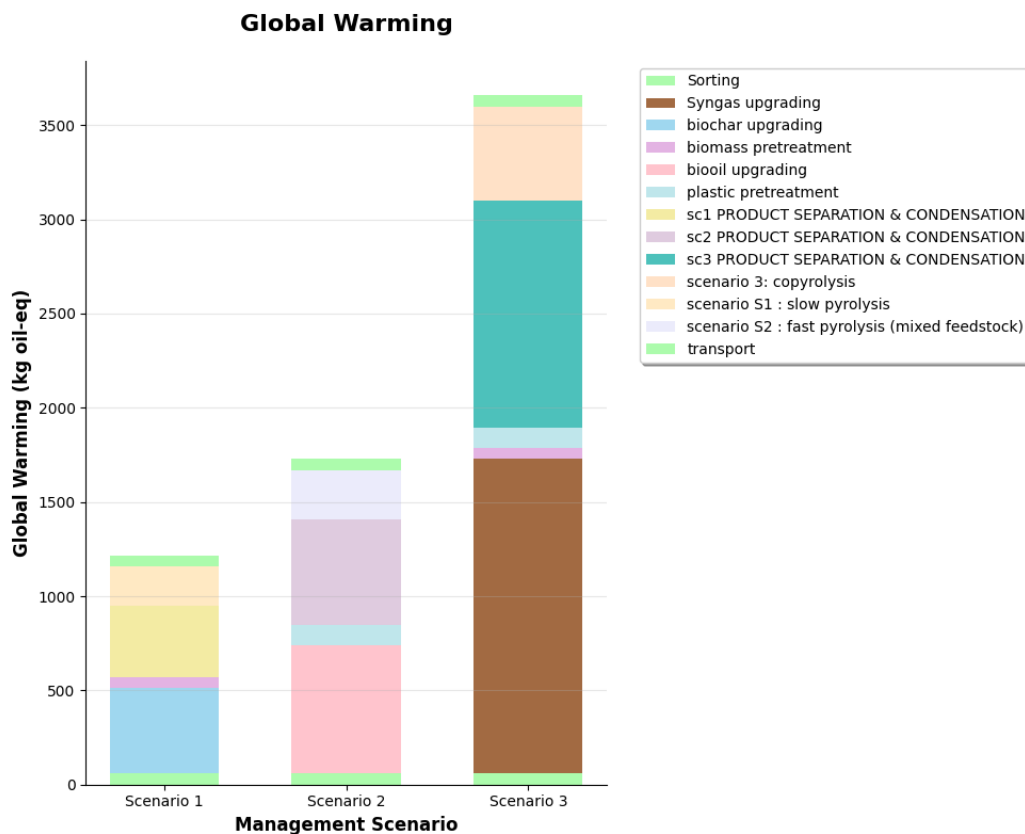


Figure 15: *Global Warming Potential across scenarios (kg CO₂-eq).*

Performance ranking. S1 < S2 < S3, with S3 showing 20-25% higher GWP than S1.

Hotspot identification.

- **S1:** Product separation and pretreatment dominate
- **S2:** Bio-oil upgrading is primary contributor (35-40%)
- **S3:** Syngas upgrading drives highest impacts (45-50%)

3.5.3 Resource Depletion

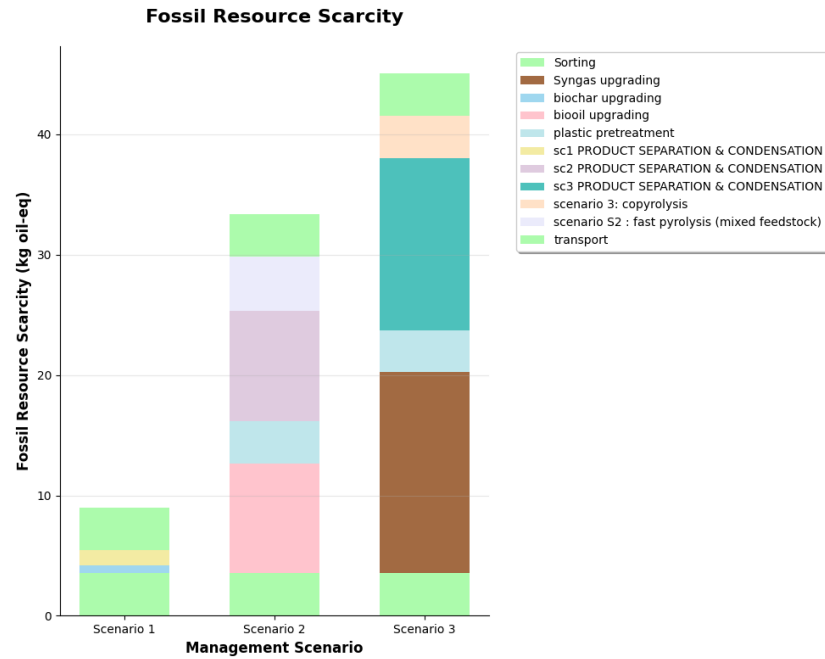


Figure 16: *Fossil Resource Scarcity by scenario.*

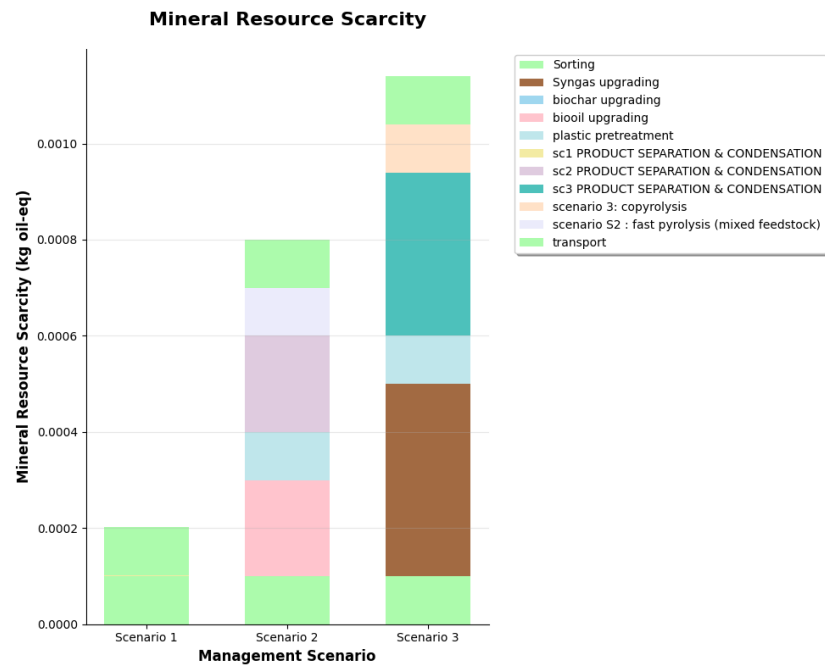


Figure 17: *Mineral Resource Scarcity by scenario.*

Fossil resource scarcity follows the pattern $S1 < S2 < S3$, driven by energy-intensive upgrading operations requiring fossil-based electricity and heat.

Mineral resource scarcity increases with process complexity: equipment-intensive upgrading trains in S2 and S3 drive higher material requirements upstream.

3.5.4 Acidification and Eutrophication

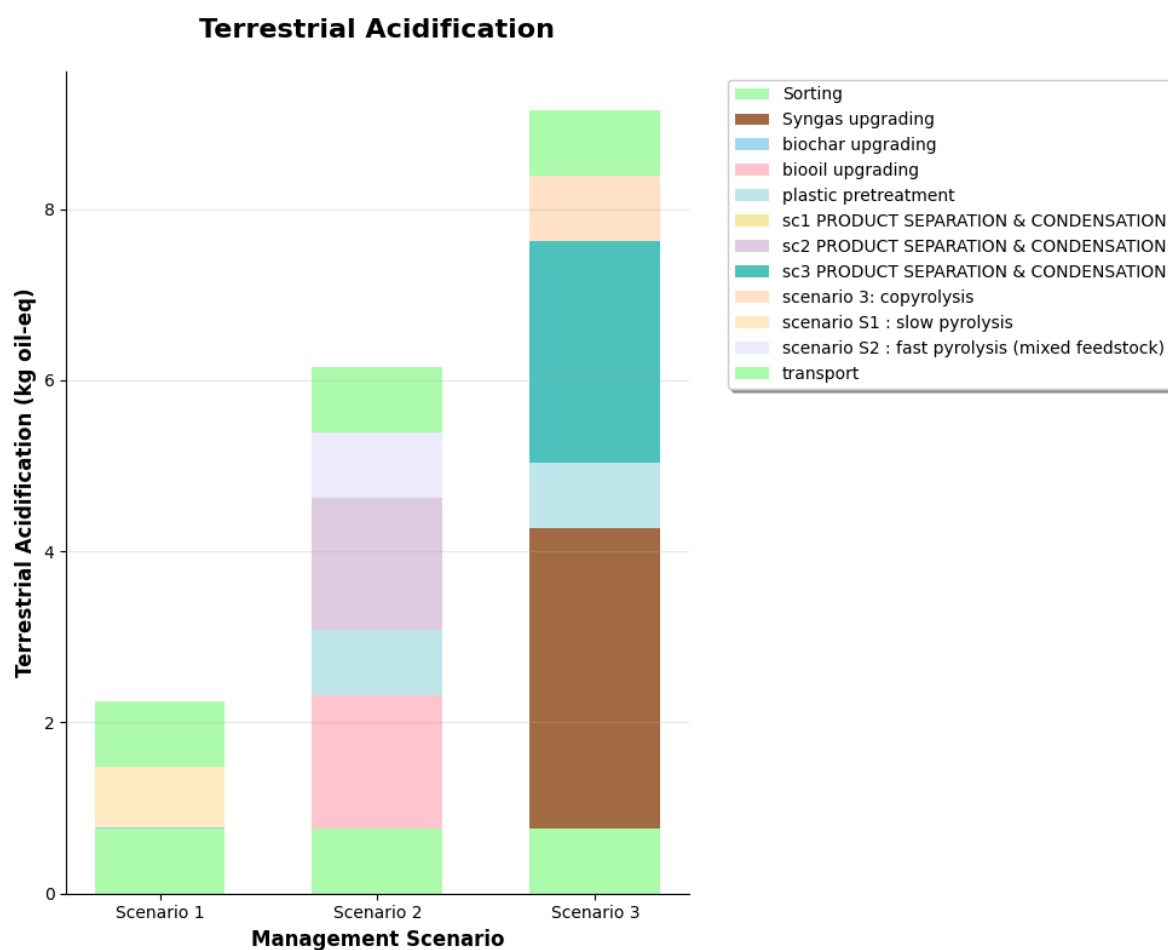


Figure 18: *Terrestrial Acidification across scenarios.*

Terrestrial acidification ranks $S1 < S2 < S3$, primarily linked to upstream SO_x , NO_x , and NH_3 emissions from electricity generation.

Eutrophication (both freshwater and marine) shows elevated impacts in S2 and S3 compared to S1, driven by upgrading and separation processes.

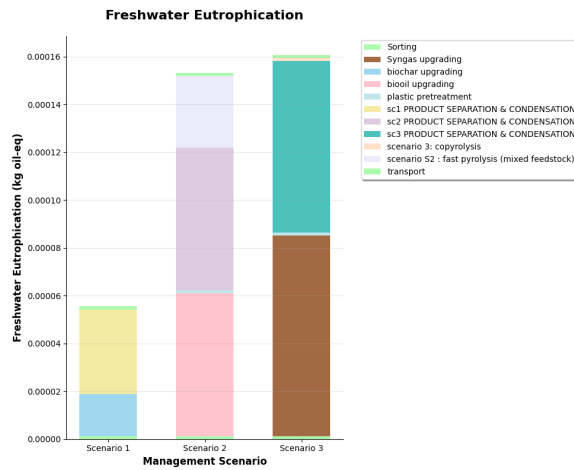


Figure 19: *Freshwater eutrophication.*

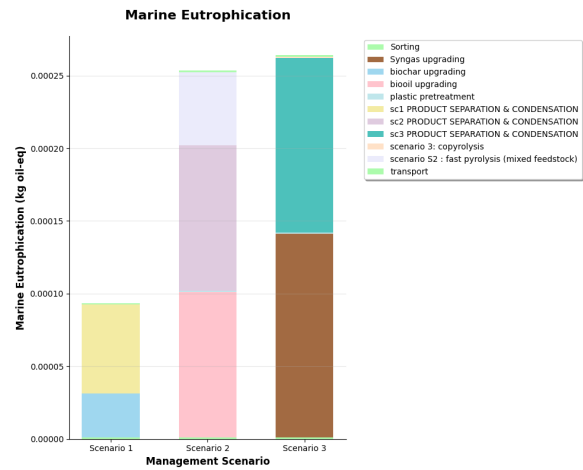


Figure 20: *Marine eutrophication.*

3.5.5 Water Consumption

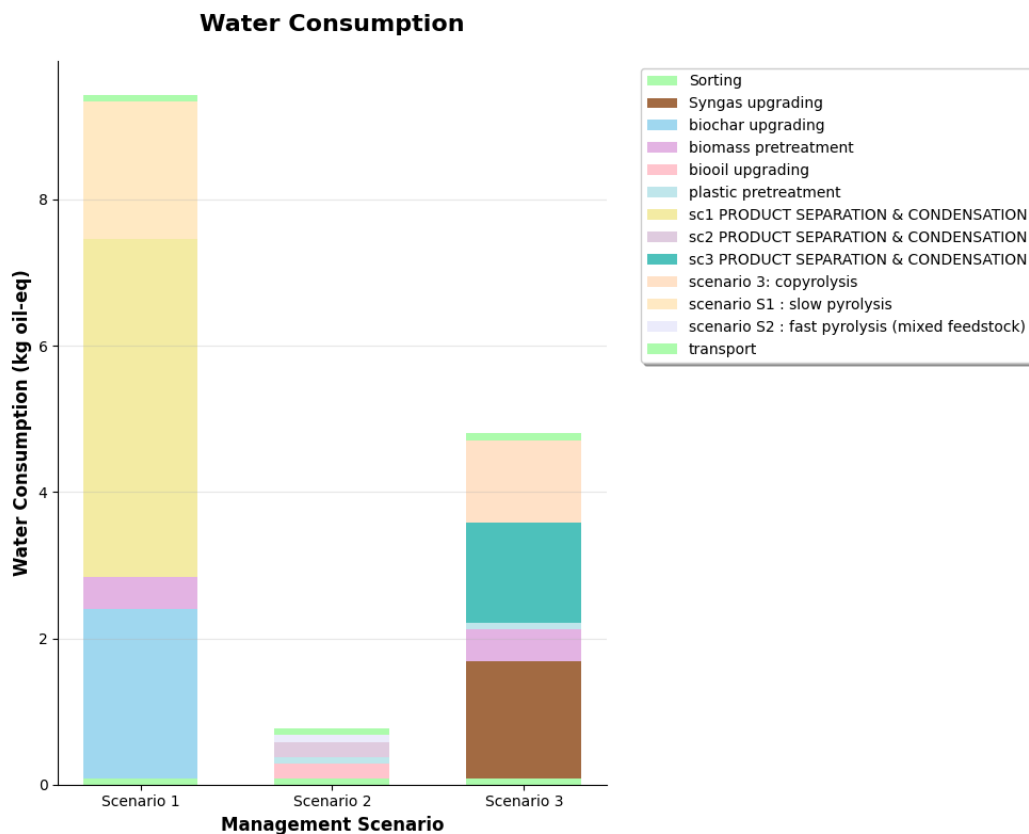


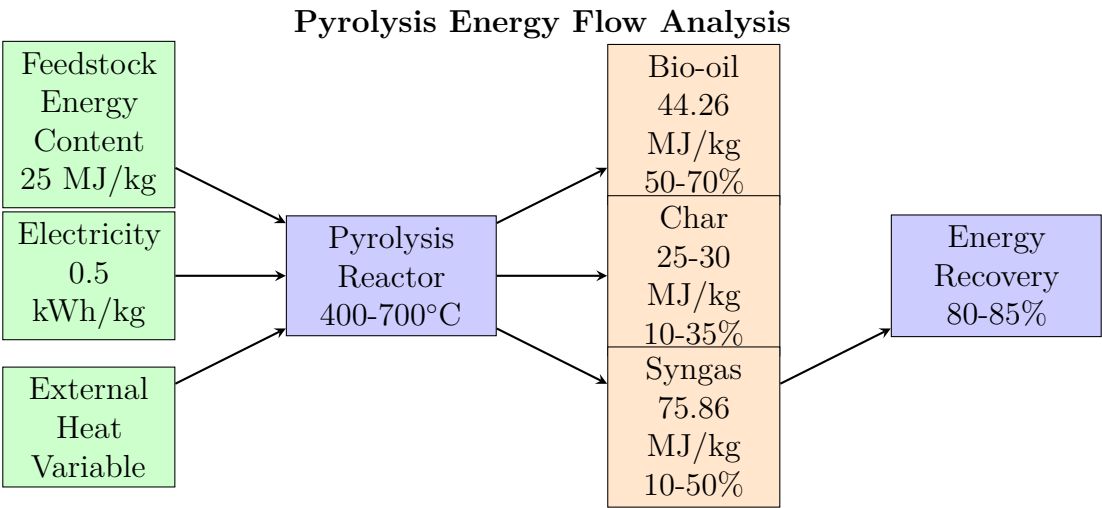
Figure 21: *Water consumption by scenario.*

S1 exhibits highest water consumption due to intensive separation processes (sc1), while S2 and S3 show substantially lower consumption. Process water recycling and heat integration offer reduction potential.

3.6 Thermodynamic Efficiency

Thermodynamic efficiency represents the cornerstone of pyrolysis LCA studies, directly influencing environmental performance across all impact categories. Energy considerations determine the net environmental benefit and economic viability of pyrolysis systems.

Energy Flow Diagram for Typical Pyrolysis System:



$$\text{Net Energy Efficiency} = [E_{\text{products}} - E_{\text{inputs}}] / E_{\text{feedstock}}$$

Figure 22: Energy flow diagram for pyrolysis process with efficiency calculations

Comparative Thermodynamic Performance:

Process Type	Electricity Consumption (kWh/kg)	Energy Efficiency (%)	GWP Impact (kg CO ₂ -eq/kg)	Fossil Fuel Depletion Reduction (%)	Overall Performance Score
Slow Pyrolysis	0.4-0.5	75-85	0.032	52-53	A
Fast Pyrolysis	0.5-0.6	80-90	0.045	45-50	A-
Microwave Pyrolysis	0.3-0.4	85-95	0.028	55-60	A+
Catalytic Pyrolysis	0.4-0.6	70-85	0.038	48-52	B+
Incineration (baseline)	0.2-0.3	25-35	0.085	0	C

Sensitivity Analysis: Impact of Electricity Grid Mix

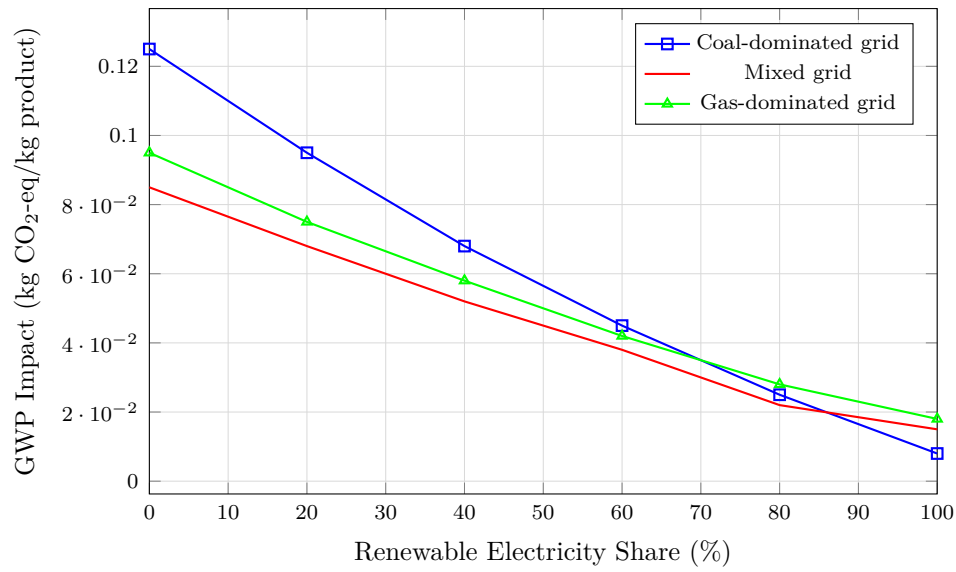


Figure 23: Sensitivity of GWP impact to electricity grid composition for pyrolysis processes

Economic-Environmental Trade-off Analysis:

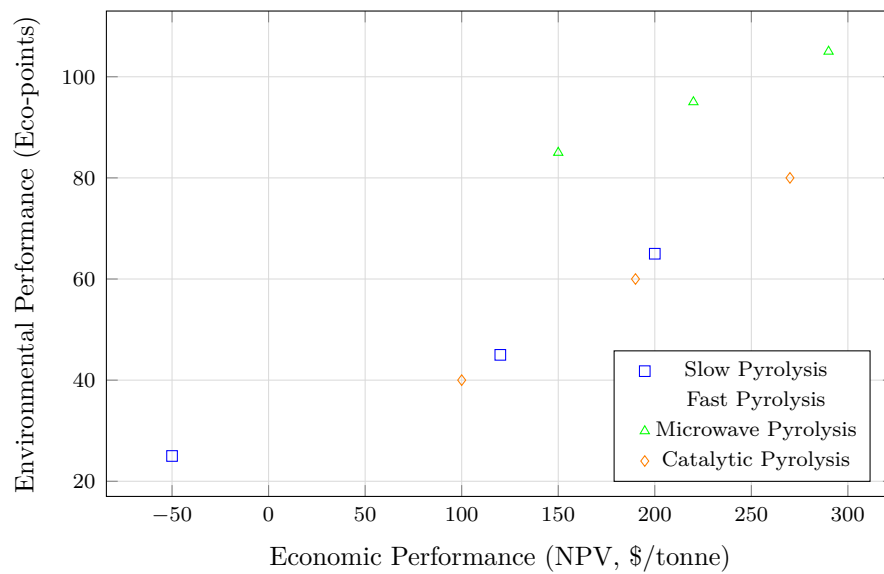


Figure 24: Economic vs. environmental performance trade-offs for different pyrolysis technologies

Key LCA Insights for Thermodynamic Efficiency:

- Energy efficiency formula: $EE = \frac{E_p - (E_f + E_i)}{0.97 \times (E_w + E_f)}$
- Electricity consumption (0.4-0.6 kWh/kg) represents largest GHG contribution
- Microwave pyrolysis shows highest conversion efficiency (80-85%)
- Grid electricity mix critically affects environmental performance
- Co-product valorization essential for positive LCA outcomes
- Heat integration can achieve near-zero net energy balance

3.7 Interpretation and Recommendations

3.7.1 Environmental Performance Summary

Table 27: *Scenario performance summary by impact category*

Impact Category	S1 Rank	S2 Rank	S3 Rank
Energy Demand (CED)	2	1	3
Climate Change (GWP)	1	2	3
Fossil Resource Scarcity	1	2	3
Water Consumption	3	1	2
Eutrophication	1	3	2
Terrestrial Acidification	1	2	3

3.7.2 Technology Selection Framework

Based on environmental priorities:

- **Climate-focused applications:** S1 recommended
- **Energy efficiency priorities:** S2 optimal
- **Water-constrained environments:** S2 or S3 preferred
- **Multi-criteria optimization:** S2 provides best balance

3.7.3 Improvement Opportunities

Universal strategies (all scenarios):

1. Heat integration and energy recovery systems
2. Renewable electricity adoption
3. Process water recycling and closed-loop systems
4. Advanced separation technologies (membranes, hybrid methods)

Scenario-specific priorities:

- **S1:** Separation system optimization, cooling efficiency
- **S2:** Bio-oil upgrading catalyst development, hydrogen sourcing
- **S3:** Fischer-Tropsch process intensification, syngas conditioning

Recommendations for future work:

- Dynamic LCA incorporating operational flexibility
 - Experimental validation of key assumptions
 - Endpoint impact assessment and monetization
 - Integration with techno-economic analysis
-

3.8 Conclusions

This LCIA provides robust environmental assessment of three pyrolysis pathways:

1. **Clear performance differentiation:** Each scenario exhibits distinct environmental profiles enabling informed technology selection
2. **Hotspot identification:** Process-level analysis identifies specific improvement targets across upgrading and separation systems
3. **Trade-off quantification:** No scenario dominates all categories; selection depends on environmental priorities and constraints
4. **Improvement pathways:** Heat integration and renewable energy adoption offer universal benefits; scenario-specific optimizations target identified hotspots

The results support evidence-based decision-making for sustainable waste-to-energy technology development and deployment, with clear guidance for both technology selection and process optimization priorities.

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